Math 239: Introduction to Combinatorics

The study of combinatorics is the study of counting things (in strange and interesting ways). Some examples:
1) How many possible poker hands are there?
2) How many ways can we choose a 3 topping pizza with 10 options for toppings?
3) How many ways can we split 8 slices of pizza between 3 people?
4) How many ways can we make change for a dollar?

Notation

$A = \{a_1, a_2, \ldots, a_n\}$ is an n-element set. All terms are different. Order doesn't matter. We denote its size by $|A|$.

Example

Let A be the set of primes less than 10, $A = \{2, 3, 5, 7\}$. Let B be the set of odd numbers less than 10, $B = \{1, 3, 5, 7, 9\}$. We have $|A| = 4, |B| = 5$.

Example/Notation

Choose a prime less than 10 and an odd number less than 10, like $(2, 1)$, $(2, 3)$, or $(7, 9)$. For this we use the notation $A \times B = \{(a, b) : a \in A, b \in B\}$, with $|A \times B| = |A||B|$. In this example, $|A \times B| = 20$.

Example/Notation

Choose a prime less than 10 or choose an odd number less than 10. The choices are $1, 2, 3, 5, 7, 9$. For this we use the notation $A \cup B = \{x : x \in A \text{ or } x \in B\}$. In this case, $|A \cup B| = 6$.

Example/Notation

Choose a number less than 10 that is both prime and an odd number. The choices are $3, 5, 7$. For this we use the notation $A \cap B = \{x : x \in A \text{ and } x \in B\}$. In this case $|A \cap B| = 3$.

Proposition

For finite sets A and B, $|A \cup B| = |A| + |B| - |A \cap B|$.

Definition

Let A be a finite set. We define a list of A to be a set of all elements of A where the order matters.

Example

Let $A = \{2, 3, 5, 7\}$ be the set of primes less than 10. The possible lists of A include $(2, 3, 5, 7)$, $(2, 3, 7, 5)$, $(2, 5, 3, 7)$, $(2, 5, 7, 3)$, $(2, 7, 3, 5)$, $(2, 7, 5, 3)$, $(3, 2, 5, 7)$ (one of the six lists starting with 3), and the twelve lists starting with 5 or 7. In this case there are 24 lists of A.

Question

Let A be a set with n elements. How many lists of A are there? Let p_n be the number of lists of A with $|A| = n$. We can partition the set of lists based on the first element of the list (i.e. all 6 lists in the above example starting with 2, all 6 lists starting with 3, etc.). There are n choices for this first element. Let $x \in A$ be the first element of a list. The remainder of the list will be chosen from the lists of $A \setminus \{x\}$. But $|A \setminus \{x\}| = n - 1$, so there are p_{n-1} lists of it to choose for the last $n - 1$ elements of our list. As there were n initial choices for x, we get $p_n = np_{n-1}$. Note $p_1 = 1$ (the only list of $\{x\}$ is (x)). Hence $p_n = np_{n-1} = n(n-1)p_{n-2} = n(n-1) \ldots p_1 = n(n-1) \ldots (1) = n!$. We have proved:

Theorem

Let A be a set with $|A| = n$. Then the number of lists of A is $n!$. ■

Definition

A subset of a set A is a collection of elements from A without repetition, order does not matter. Note that a subset may be empty, or all of A.

Example
Let $A = \{2,3,5,7\}$. The set of subsets of A is $\{\emptyset\{2\}, \{3\}, \{5\}, \{7\}, \{2,3\}, \{2,5\}, \{2,7\}, \{3,5\}, \{3,7\}, \{5,7\}, \{2,3,5\}, \{2,3,7\}, \{2,5,7\}, \{3,5,7\}, A\}$.

We see for each $x \in A$ a subset either contains x, or it does not contain x. This gives us two choices for all $x \in A$, either it is in the subset or not. This tells us:

Theorem
Let A be a finite set with $|A| = n$. Then there are 2^n subsets of A. ■

Definition
Let A be a set of size n. A partial list of size k is an ordered list $(a_1, \dots, a_k)$ with $a_i \in A$ with all a_i distinct.

Example
Let $A = \{2,3,5,7\}$. Let $k = 2$. The partial lists of length 2 of A are $(2,3)$, $(3,2)$, $(2,5)$, $(5,2)$, $(2,7)$, $(7,2)$, $(3,5)$, $(5,3)$, $(3,7)$, $(7,3)$, $(5,7)$, $(7,5)$.

Question
For any n and k, how many partial lists of length k are there of $\{1,2,\dots,n\}$. Notice that in the previous example there are 4 choices for the first element of the list. After we have chosen the first element, we have 3 choices for the second element. After this, we are done. This gives the number of partial lists in the previous example as $4 \cdot 3$.

In general, for $|A| = n$ and $k \leq n$, we have n choices for the first element, $n-1$ choices for the second, up to $n-k+1$ choices for the kth element, so the number of partial lists of length k is:

Theorem
The number of partial lists of length k of a finite set A of size n is $n(n-1)\dots(n-k+1)$
$= \dfrac{n(n-1)\dots(n-k+1)(n-k)\dots(2)(1)}{(n-k)(n-k-1)\dots(2)(1)} = \dfrac{n!}{(n-k)!}$. ■

Example
Find all of the subsets of size 2 of $\{2,3,5,7\}$: $\{2,3\}$, $\{2,5\}$, $\{2,7\}$, $\{3,5\}$, $\{3,7\}$, $\{5,7\}$.

For every subset of length k, there are $k!$ orderings of the elements to get a partial list of length k (we already proved this, just for sets and lists, and not subsets and partial lists). This is true for every subset. This gives us that $k!$(number of subsets of size k of set of size n) = (number of partial lists of length k of set of size n), in other words:

Theorem
Let A be a finite set of size n and let $k \in \mathbb{N}$ such that $k \leq n$. The number of subsets of A of size k is $\frac{n!}{(n-k)!k!} = \binom{n}{k}$. ■

1.1.3 Meaning vs Algebra

There are many examples in combinatorics where we can manipulate the algebra to get a result or relationship. This will tell you that something is true, but not why it is true. We wish to find relationships between things we know how to count and things we want to count. Such an argument is called a combinatorial proof.

Example
We want to show $\binom{n}{k} = \binom{n}{n-k}$ combinatorially. We see that $\binom{n}{k}$ is the number of subsets of size k of $\{1, \ldots, n\}$. Let $S = \{a_1, \ldots, a_k\}$ be a subset of size k of $\{1, \ldots, n\}$. Notice $S^c = \{1, \ldots, k\} \setminus S$ is a subset of size $n-k$. We see every subset A of size k corresponds to a unique subset A^c of size $n-k$. Furthermore, every subset of size $n-k$ can be constructed in this way. This tells us that the number of subsets of size k is the same as the number of subsets of size $n-k$, that is $\binom{n}{k} = \binom{n}{n-k}$.

Definition
Let A and B be sets. A map $f : A \to B$ is said to be surjective (onto) if for all $b \in B$ there exists (at least one) $a \in B$ with $f(a) = b$.

Definition
A map $f : A \to B$ is said to be injective if $f(a) = f(b)$ implies $a = b$.

Definition
A map $f : A \to B$ is called bijective if it is both injective and surjective.

Example
$f : \{1, 2, 3\} \to \{2, 3, 5\}$ by $f(1) = 2$, $f(2) = 3$, $f(3) = 5$. This is a bijection. Notice that bijections can be inverted, in this case: $f^{-1}(2) = 1$, $f^{-1}(3) = 2$, $f^{-1}(5) = 3$.

Observation
If there is a surjection $f : A \to B$ then $|A| \geq |B|$, and if there is an injection $f : A \to B$ then $|A| \leq |B|$, and therefore if there is a bijection $f : A \to B$ then $|A| = |B|$.

Notation
If there is a bijection $f : A \to B$, we say (a character I don't know, I use isomorphisms) $A \cong B$.

Example
Show $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. Let $B(n, k)$ be the set of subsets of $\{1, \ldots, n\}$ of size k. We know $|B(n, k)| = \binom{n}{k}$, and $|B(n-1, k-1)| + |B(n-1, k)| = \binom{n-1}{k-1} + \binom{n-1}{k}$. Our goal is to find a bijection from $B(n, k)$ to $B(n-1, k-1) \cup B(n-1, k)$. Notice $B(n-1, k-1)$ and $B(n-1, k)$ are disjoint, which gives $|B(n-1, k-1) \cup B(n-1, k)| = |B(n-1, k-1)| + |B(n-1, k)|$. We will construct a map $f : B(n, k) \to B(n-1, k-1) \cup B(n-1, k)$ by $f(\{a_1, \ldots, a_k\}) = \{a_1, \ldots, a_k\} \setminus \{n\}$. If $n \notin \{a_1, \ldots, a_k\}$ we see $f(\{a_1, \ldots, a_k\}) = \{a_1, \ldots, a_k\} \in B(n-1, k)$. If $n \in \{a_1, \ldots, a_k\}$, then $f(\{a_1, \ldots, a_k\})$ is of size $k-1$, using elements from $\{1, \ldots, n-1\}$. This gives $f(\{a_1, \ldots, a_k\}) \in B(n-1, k-1)$. We must prove f is a bijection. Notice if $n \notin \{a_1, \ldots, a_k\}$, this map is injective and surjective (since it's the identity). If $n \in \{a_1, \ldots, a_k\}$, say $a_k = n$ without loss of generality, then we can write down the inverse $f^{-1}(a_1, \ldots, a_{k-1}) = \{a_1, \ldots, a_k\}$. Thus in both cases f is a bijection.

Definition
Let $n \geq 0, t \geq 1$. A multiset of size n with t types is a list $(a_1, \ldots, a_t)$ where $a_1 + \cdots + a_t = n$.

Definition
A person has 10 pets, of types fish, cats, and dogs. Here $n = 10, t = 3$. Then the list $(7, 2, 1)$ could represent 7 fish, 2 cats, 1 dog.

Theorem
The number of multisets of size n of type t is $\binom{n+t-1}{t-1}$.
Proof:

We will do this by a bijective method. Let A be the set of subsets of size $t-1$ of $\{1,\ldots,n+t-1\}$. Let B be our set of multisets of size n with t types. For example, consider the subset in A given by writing $n+t-1$ circles, with $t-1$ circles crossed off. This partitions the list of circles into groups, corresponding to the respective types of the multiset. This map is a bijection from A to B, giving $|A|=|B|$. ■

Formal Power Series

In Calculus 2, we considered power series. The key difference in this course is that we don't care about convergence. What we care about are the coefficients and the information they carry.

Definition
A formal power series is a series $A(x)=\sum_{n=0}^{\infty} a_n x^n$. Often, we restrict the a_n to be positive integer and typically represent the size of some set we are interested in.

Example
Let a_n be the number of lists of $\{1,\ldots,n\}$. In this case, $a_n = n!$. We create the power series $A(x)=\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0} n!x^n$. This only converges at $x=0$, but we don't care.

Definition
We can (and do) add and multiply formal power series in the obvious way: $\sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty}(a_n+b_n)x^n$, and $\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = (a_0+a_1x+a_2x^2+\ldots)(b_0+b_1x+b_2x^2+\ldots) = a_0b_0+(a_0b_1+a_1b_0)x+(a_0b_2+a_1b_1+a_2b_0)x^2+\cdots = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} a_k b_{n-k}\right)x^n$.

Remark
In this course, we often call x an indeterminate, not a variable. Roughly, this is because x will not take values we will evaluate it at.

We often will manipulate formal power series algebraically, disregarding convergence.

Example
Let $A(x)=\sum_{n=0}^{\infty} x^n$. Notice that $xA(x)=\sum_{n=1}^{\infty} x^n$. This gives $A(x)-xA(x)=1$, or equivalently $(1-x)A(x)=1$, or $A(x)=1/(1-x)$ or $1-x=1/A(x)$, neither of which are formal power series.

Theorem (Binomial Theorem)
Let $n\geq 0$ be fixed. Let a_k be the number of subsets of $\{1,\ldots,n\}$ of size k. We have $\sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{n}\binom{n}{k}x^k=(1+x)^n$ (the first equality holds since if $k>n$ then $a_k=0$).
Proof:
Let $\mathcal{P}(n)$ be the set of all subsets of $\{1,\ldots,n\}$. Let $\mathcal{B}(n)$ be the set of all length n binary strings. We define a bijection from $\mathcal{P}(n)\to\mathcal{B}(n)$ in the obvious (to Alex because he's seen this 182931232142893712 times) way: give a binary string a 1 in the ith index if the subset of $\{1,\ldots,n\}$ contains i. We have $\sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{n}\sum_{\{c_1,\ldots,c_k\}\in\mathcal{P}(n)} x^k = \sum_{k=0}^{n}\sum_{(b_1,\ldots,b_k)\in\mathcal{B}(n)} x^{b_1+\cdots+b_k} = \prod_{i=1}^{n}\sum_{b_i\in\{0,1\}} x^{b_i} = (1+x)^n$. ■

Example
Let's show $\binom{m+n}{k}=\sum_{j=0}^{k}\binom{m}{j}\binom{n}{k-j}$. By the above theorem, we have $(1+x)^{m+n}=\sum_{k=0}^{m+n}\binom{m+n}{k}x^k$. Furthermore, we have $(1+x)^{m+n}=(1+x)^m(1+x)^n=\sum_{j=0}^{m}\binom{m}{j}x^j\sum_{i=0}^{n}\binom{n}{i}x^n$. Multiplying out, this gives us $(1+x)^{m+n}=\sum_{j=0}^{m}\sum_{i=0}^{n}\binom{m}{j}\binom{n}{i}x^{j+i}=\sum_{k=0}^{m+n}\sum_{i+j=k}\binom{m}{j}\binom{n}{i}x^{j+i}=\sum_{k=0}^{m+n}\sum_{i+j=k}\binom{m}{j}\binom{n}{k-j}x^k=\sum_{k=0}^{m+n}\sum_{j=0}^{k}\binom{m}{j}\binom{n}{k-j}x^k$. Thus, looking coefficient wise at our series for $(1+x)^{m+n}$ we have $\binom{m+n}{k}=\sum_{j=0}^{k}\binom{m}{j}\binom{n}{k-j}$.

Notation
Let $[x^k]A(k)$ denote the kth coefficient of a power series A.

Generating Series

Definition
Let $g_0, g_1, g_2, \ldots$ be a sequence of numbers that we care about, and that encode some information. For example, g_n is the number of binary strings of length n, or g_n is the number of partial lists of length n of $\{1, 2, \ldots, 100\}$. We define the generating series of the sequence $(g_n)_{n=1}^{\infty}$ as the formal power series $G(x) = \sum_{n=0}^{\infty} g_n x^n$.

We often generate these series using a weight function:

Definition
Let A be a set. We say $w : A \to \mathbb{N}$ (our convention is that $\mathbb{N}$ includes 0) is a weight function of A if $A_n := w^{-1}(n)$ is a finite set for all n.

Example
Let A be the set of all binary strings of any length. The function $w : A \to \mathbb{N}$ that gives the length of the string is a good choice for a weight function (a bad one would be the number of 1s in the string, since preimages of positive integers are infinite), where $|A_n| = 2^n$.

Example
Let A be the set of all subsets of $\{1, \ldots, 10\}$. Define $w : A \to \mathbb{N}$ by $w(\{a_1, \ldots, a_k\}) = k$. Then $A_0 = \{\emptyset\}$, $A_1 = \{\{1\}, \ldots, \{10\}\}$, and $A_n = \emptyset$ for $n \geq 11$.

Definition
We define the generating series for A with respect to a weight function w for A as $\Phi_A^w(x) = \sum_{a \in A} x^{w(a)}$.

Example
In the above example, $\Phi_A^w(x) = x^{w(\emptyset)} + x^{w(\{1\})} + x^{w(\{2\})} + \cdots + x^{w(\{1,\ldots,10\})} = x^0 + x^1 + x^1 + \cdots + x^{10} = x^0 + \binom{10}{1}x + \binom{10}{2}x^2 + \cdots + \binom{10}{9}x^9 + \binom{10}{10}x^{10} = \sum_{n=0}^{10} |A_n| x^n = \sum_{n=0}^{\infty} |A_n| x^n = \sum_{n=0}^{\infty} \binom{10}{n} x^n$. Note that this follows since $|A_n|$ is the number of subsets of size n taken from $\{1, \ldots, 10\}$, we have $|A_n| = \binom{10}{n}$.

Implicitly in this example we used:
Theorem
Let A be a set and w a weight function of A. Then $\Phi_A^w(x) = \sum_{a \in A} x^{w(a)} = \sum_{n=0}^{\infty} |A_n| x^n$.
Proof:
We have

$$\sum_{a \in A} x^{w(a)} = \sum_{a \in A_0 \cup A_1 \cup \ldots} x^{w(a)} = \sum_{n=0}^{\infty} \sum_{x \in A_n} x^{w(a)} \overbrace{=}^{\text{def. of } A_n} \sum_{n=0}^{\infty} \sum_{a \in A_n} x^n = \sum_{n=0}^{\infty} x^n \sum_{a \in A_n} 1 = \sum_{n=0}^{\infty} |A_n| x^n \quad \blacksquare$$

Theorem (Binomial Series)
Let g_n be the number of multisets of n of t types: $g_n = \binom{n+t-1}{t-1}$. We have $\sum_{n=0}^{\infty} g_n x^n = \frac{1}{(1-x)^t}$.
Proof:
Fix t. We want A_n to be all the multisets of n with t types. Therefore, define $w((a_1, \ldots, a_t)) = a_1 + \cdots + a_t$ for the set $A = \mathbb{N}^t$. This gives $\sum_{n=0}^{\infty} \binom{n+t-1}{t-1} x^n = \sum_{a \in A} x^{w(a)} = \sum_{n=0}^{\infty} |A_n| x^n$. But $\sum_{a \in A} x^{w(a)} = \sum_{(a_1, \ldots, a_t) \in A} x^{a_1 + \cdots + a_t} = \sum_{(a_1, \ldots, a_t) \in A} x^{a_1} \ldots x^{a_t} = (\sum_{a_1=0}^{\infty} x^{a_1}) \ldots (\sum_{a_t=0}^{\infty} x^{a_t}) = (\frac{1}{1-x}) \ldots (\frac{1}{1-x}) = (\frac{1}{1-x})^t$. $\blacksquare$

Formal Power Series (Part 2)

Recall from Calculus 2 we have $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$. Disregarding convergence, we notice that $1 = (1-x)(1+x+x^2+\dots)$, where both $(1-x)$ and $(1+x+x^2+\dots)$ are formal power series.

Definition

Let $A(x)$ and $B(x)$ be formal power series. If $A(x)B(x) = 1$, then we say $A(x)$ is an inverse of $B(x)$ (and vice versa).

Example

Let $A(x) = 1 - x - x^2$. Let's find an inverse (if it exists). Write $B(x) = b_0 + b_1 x + \dots$. This gives $A(x)B(x) = (1-x-x^2)(b_0 + b_1 x + \dots) = b_0 + (b_1 - b_0)x + (b_2 - b_1 - b_0)x^2 + (b_3 - b_2 - b_1)x^3 + \dots$. Notice $[x^0]1 = 1$, and $[x^0]A(x)B(x) = b_0$, so we get $b_0 = 1$. Similarly, notice for all $k \neq 0$, $[x^k]1 = 0$. But $[x^1]A(x)B(x) = b_1 - b_0 = b_1 - 1$, so we get $b_1 = 1$. For general $n \geq 2$, we have $[x^n]1 = 0$, and $[x^n]A(x)B(x) = b_n - b_{n-1} - b_{n-2}$, so $b_n = b_{n-1} + b_{n-2}$. This is the Fibonacci sequence. Therefore, we have found an inverse $B(x)$ for $A(x)$, specifically $B(x) = 1 + x + 2x^2 + 3x^3 + 5x^4 + \dots$.

Example

We'll show $A(x) = x + x^2$ does not have an inverse. Assume it does, say $B(x) = b_0 + b_1 x + b_2 x^2 + \dots$. As before, we will consider $A(x)B(x)$ and match coefficients. For $n = 0$, note that $A(x)B(x) = b_0 x + (b_1 + b_0)x^2 + \dots$, so for $n = 0$, $[x^0]A(x)B(x) = 0 \neq 1 = [x^0]1$, so we cannot make an inverse.

Theorem

Let $A(x) = \sum_{k=0}^{\infty} a_k x^k$ be a formal power series. Then there exists an inverse $B(x)$ of $A(x)$ if and only if $a_0 \neq 0$.

Proof:

Formalize the above examples. ■

Definition

Let $A(x)$ and $B(x)$ be formal power series. We define the composition as $A(B(x)) = a_0 + a_1 B(x) + a_2 B(x)^2 + \dots$ if it exists.

Example

Let $A(x) = 1 + x + x^2 + \dots$, $B(x) = x + x^2$. So $A(B(x)) = 1 + (x+x^2) + (x+x^2)^2 + (x+x^2)^3 = 1 + x + 2x^2 + 3x^3 + 5x^4 + 8x^5$. This is the same series as before when we found the inverse for $1 - x - x^2$. Why? (Exercise)

Example

Let $A(x)$ be as before, and $B(x) = 1 + x$. Then $A(B(x)) = 1 + (1+x) + (1+x^2) + \dots$, where the coefficient on each term will diverge, so this composition does not exist.

Theorem

Let $A(x)$ and $B(x) = b_0 + b_1 x + \dots$ be formal power series. If $b_0 = 0$, then $A(B(x))$ is well defined.

Proof:

Assume $b_0 = 0$. We can write $B(x) = xC(x)$. Hence $A(B(x)) = A(xC(x)) = a_0 + a_1 x C(x) + a_2 x^2 C(x)^2 + \dots$. Notice $[x^n]A(xC(x)) = [x^n] \sum_{k=0}^{n} a_k x^k C(x)^k$. This sum is finite, hence all coefficients are finite. ■

Definition

We say a power series $A(x)$ is rational if there exist polynomials $P(x)$ and $Q(x)$ such that $A(x) = Q(x)A(x)$ for $Q(x)$ nonzero.

Example

Let $A(x)$ be the power series whose coefficients are the Fibonacci numbers. Note $A(x) = \frac{1}{1-x-x^2}$, hence $A(x)$ is rational.

Sum & Product Lemma

Theorem (Sum Lemma)
Let A and B be disjoint sets and let $S = A \cup B$. Let $w : S \to \mathbb{N}$ be a weight function (and hence a weight function on A and B). Then $\Phi_S^w(x) = \Phi_A^w(x) + \Phi_B^w(x)$.
Proof:

$$\Phi_S^w = \sum_{s\in S} x^{w(s)} = \sum_{s\in A\cup B} x^{w(s)} = \sum_{s\in A \text{ or } s\in B} x^{w(s)} = \sum_{s\in A} x^{w(s)} + \sum_{s\in B} x^{w(s)} = \Phi_A^w(x) + \Phi_B^w(x) \quad \blacksquare$$

Remark
Note that $\Phi_A^w(x) = \sum_{n=0}^{\infty} |A_n| x^n = \sum_{s\in S} x^{w(s)}$, where the second term is what we care about combinatorially, and the third term is what we care about for proofs.

Example
Let A be all the subsets of $\{1,\ldots,n\}$ that contain n, and B be all the subsets of $\{1,\ldots,n\}$ do not contain n. We have $A \cap B = \emptyset$, and $S = A \cup B$ is the set of all subsets of $\{1,\ldots,n\}$. Let $w(\{c_1,\ldots,c_k\}) = |\{c_1,\ldots,c_k\}|$ be the weight function that assigns the size of each subset. Then $\Phi_S^w(x) = \sum_{s\in S} x^{w(s)} = \sum_{k=0}^{\infty} |S_k| x^k$. Here S_k is all subsets of $\{1,\ldots,n\}$ of size k, so $|S_k| = \binom{n}{k}$, so $\Phi_S^w(x) = \sum_{k=0}^{\infty} \binom{n}{k} x^k = \sum_{k=0}^{n} \binom{n}{k} x^k$. Let A_k be the set of all subsets of A of size k. There is a natural bijection from A_k to $\tilde{A}_k$, the subsets of $\{1,\ldots,n-1\}$ of size $k-1$ (delete n). This gives us $\Phi_A^w(x) = \sum_{k=0}^{\infty} |A_k| = \sum_{k=0}^{\infty} |\tilde{A}_k| x^k = \sum_{k=0}^{\infty} \binom{n-1}{k-1} x^k$. Here B_k is the subsets of size k from $\{1,\ldots,n\}$, which can equivalently be thought of as the subsets of size k from $\{1,\ldots,n-1\}$. This gives $\Phi_B^w(x) = \sum_{k=0}^{\infty} |B_k| = \sum_{k=0}^{\infty} \binom{n-1}{k} x^k$. By the sum lemma, we have $\Phi_S^w(x) = \Phi_A^w(x) + \Phi_B^w(x)$, or $\sum_{k=0}^{\infty} \binom{n}{k} x^k = \sum_{k=0}^{\infty} \binom{n-1}{k-1} x^k + \sum_{k=0}^{\infty} \binom{n-1}{k} x^k$, so $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Theorem (Product Lemma)
Let A be a set with weight function w, and B be a set with weight function v, and let $S = A \times B$. We define a weight function on S as $\sigma(s) = (w\times v)(s) = (w \times v)(a,b) = w(a) + v(b)$. Then $\Phi_S^\sigma(x) = \Phi_{A\times B}^{w\times v} = \Phi_A^w(x)\Phi_B^v(x)$.
Proof:

$$\Phi_S^\sigma(x) = \sum_{s\in S} x^{\sigma(s)} = \sum_{(a,b)\in A\times B} x^{w(a)+v(b)} = \sum_{a\in A}\sum_{b\in B} x^{w(a)} x^{v(b)} = \sum_{a\in A} x^{w(a)} \sum_{b\in B} x^{v(b)} = \Phi_A^w(x)\Phi_B^v(x) \quad \blacksquare$$

Example
Let $A = \{1,2,3,4,5,6\}$ be the possible faces of a d6. Let $w(a) = a$. So $\Phi_A^w(x) = x + x^2 + x^3 + x^4 + x^5 + x^6$. Let $S = A \times A$ be the 36 possible ways you can roll a pair of dice. In this case, $\Phi_{A\times A}(x) = \Phi_A(x)\Phi_A(x) = (\Phi_A(x))^2 = (x + x^2 + \cdots + x^6)^2 = x^2 + 2x^3 + 3x^4 + 4x^5 + 5x^6 + 6x^7 + \cdots + x^{12}$. Thus there are 4 possible ways the sum of two dice is equal to 5 (coefficient of x^5 term), and similarly for other dice.

We often apply the Product Lemma by induction to higher Cartesian products.

Definition
We define $A^* = \bigcup_{k=0}^{\infty} A^k$.

Example
Let $A = \{0,1\}$. Then $A^* = A^0 \cup A_1 \cup A_2 \cup \cdots = \{(), (0), (1), (0,0), (0,1), (1,0), (1,1), \ldots\}$ is the set of all binary strings.

Definition
We define $w^* : A^* \to \mathbb{N}$ by the property if $a \in A^k$ then $w^*(a) = w^k(a)$.

Example
Let $A = \{1, 2\}$ and $w : A \to \mathbb{N}$ by $w(1) = 1$, $w(2) = 2$. Then $(1, 2, 2, 1, 1) \in A^*$, so $w^*(1, 2, 2, 1, 1) = w(1) + w(2) + w(2) + w(1) + w(1) = 7$.

There will be situations where w^* is not a weight function. For example, if there exists $a \in A$ with $w(a) = 0$. Then $w^*(a) = 0, w^*(a, a) = 0, w^*(a, \ldots, a) = 0$ for any number of inputs, so its preimage of 0 is not finite. This is the only way this construction will not work:

Theorem (String Lemma)
Let A be a set and $w : A \to \mathbb{N}$ a weight function on A such that $w(a) \neq 0$ for all $a \in A$. Then w^* is a weight function and $\Phi^{w^*}_{A^*}(x) = \frac{1}{1-\Phi^w_A(x)}$.
Proof:

$$\Phi^{w^*}_{A^*}(x) = \Phi^{w^*}_{\bigcup_{k=0}^{\infty} A_k}(x) \overbrace{=}^{\text{sum lemma}} \sum_{k=0}^{\infty} \Phi^{w^*}_{A^k}(x) = \sum_{k=0}^{\infty} \Phi^{w^k}_{A_k}(x)$$
$$\overbrace{=}^{\text{prod. lemma}} \sum_{k=0}^{\infty} (\Phi^w_A(x))^k \overbrace{=}^{\text{geom. series}} \frac{1}{1 - \Phi^w_A(x)}$$

where the sum lemma and product hold by finite induction here since $w(a) \neq 0$, so $w^*(a_1) \geq 1$, $w^*(a_2) \geq 2$, in general $w^*(a_i) \geq i$ for $a_i \in A^i$, so the sum only "sees" finitely many terms each index. (If you are reading my notes, ignore this last part, it was my explanation of a rigour thing that doesn't matter for the course). ■

Example
Let $A = \{1, 2\}$ and $w : A \to \mathbb{N}$ be as above with $w(1) = 1, w(2) = 2$. Then $\Phi^w_A(x) = x^{w(1)} + x^{w(2)} = x + x^2$. Then $\Phi^{w^*}_{A^*}(x) = \frac{1}{1-\Phi^w_A(x)} = \frac{1}{1-x-x^2} = 1 + x + 2x^2 + 3x^3 + 5x^4 + \ldots$ is the Fibonacci sequence. We notice that in this case that $[x^4]\Phi^{w^*}_{A^*}(x) = 5$, corresponding to the strings $(1, 1, 1, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (2, 2)$ that add up to 4.

Compositions

Definition
A composition is a list of positive integers $(a_1, \ldots, a_k)$. The entries a_i are called the composition's parts, its length is k, and its size is the sum of its parts.

Example
The compositions of size 4 include $(1, 1, 1, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (2, 2)$ from the last example, there are three more: $(4), (3, 1), (1, 3)$.

Let $\mathcal{C}$ be the set of all compositions. What is $\Phi^{\text{size}}_{\mathcal{C}}(x)$? We know that there are 8 compositions of size 4, for example. Let $\mathcal{C}_n$ be all the compositions of length n. For instance, $\mathcal{C}_1 = \mathbb{N}^1$, $\mathcal{C}_2 = \mathbb{N}^2$, and in general $\mathcal{C}_n = \mathbb{N}^n$. We have $\Phi^{\text{size}}_{\mathcal{C}_1}(x) = x + x^2 + x^3 + \cdots = \frac{x}{1-x}$. This allows us to write

$$\Phi^{\text{size}}_{\mathcal{C}}(x) = \sum_{k=0}^{\infty} \Phi^{\text{size}}_{\mathcal{C}_k}(x) = \sum_{k=0}^{\infty} \Phi^{\text{size}}_{\mathcal{C}_1}(x)^k = \frac{1}{1 - \Phi^{\text{size}}_{\mathcal{C}_1}(x)} = \frac{1}{1 - \frac{x}{1-x}} = \frac{1-x}{1-2x} = 1 + \sum_{k=1}^{\infty} 2^{k-1} x^k$$

From this we conclude that the number of compositions of size k is 2^{k-1} if $k \geq 1$, and 1 if $k = 0$.

Example
Let g_n be the number of compositions of n into 2 or more parts using $1, 3, 7$. Let's find a closed form for $\sum_{n=0}^{\infty} g_n x^n$. Let A be the set $\{1, 3, 7\}$. We will only be looking at $A^2, A^3, A^4, \dots$. We see $\Phi_A(x) = x + x^3 + x^7$. The generating series we are interested in is $\sum_{n=2}^{\infty} \Phi_{A_n}(x) = \sum_{n=2}^{\infty} \Phi_A(x)^n$. Note $\sum_{n=2}^{\infty} y^n = y^2 \sum_{n=0}^{\infty} y^n = \frac{y^2}{1-y}$. This allows us to find the generating series we want as $\sum_{n=2}^{\infty} (\Phi_A(x))^n = \frac{\Phi_A(x)^2}{1-\Phi_A(x)} = \frac{(x+x^3+x^7)^2}{1-x-x^3-x^7}$.

Example
Find the number of partitions of n, using only odd numbers, into an odd number of parts. Some examples include $(1), (3), (5), (7), \dots,\ (1,1,1), (1,1,3), (1,3,1), (3,1,1), \dots,$ and $(1,1,1,1,1), \dots$. As before, it is useful to determine a generating series into exactly 1 part, then apply the product lemma. In this case, A is the set of odd numbers. So

$$\Phi_A(x) = \sum_{a \text{ odd}} x^{w(a)} = \sum_{a \text{ odd}} x^a = x+x^3+x^5+\cdots = x(1+x^2+x^4+x^6+\dots) = x(1+(x^2)^1+(x^2)^2+(x^3)^2+\dots)$$

This gives us that $\Phi_A(x) = x(1 + (x^2) + (x^2)^2 + \dots) = \frac{x}{1-x^2}$. To find the generating series into an odd number of parts, we use

$$\sum_{n \text{ odd}} \Phi_{A^n}(x) = \sum_{n \text{ odd}} \Phi_A(x)^n = \frac{\Phi_A(x)}{1 - \Phi_A(x)^2} = \frac{\frac{x}{1-x^2}}{1 - (\frac{x}{1-x^2})^2} = \frac{x - x^3}{1 - 3x^2 + x^4}$$
$$= x + 2x^3 + 5x^5 + 13x^7 + 34x^9 + \dots$$

We note that we can check this answer by verifying that if any coefficients are negative or any coefficients do not match up with an exhaustive search (in the small case), we have made a mistake. So from the series there should be 5 compositions of 5 into an odd number of parts, and there are: $(1,1,1,1,1), (1,1,3), (1,3,1), (3,1,1), (5)$.

Binary Strings

Definition
A binary string is a string of the form $a_1 a_2 \dots a_n$, where $a_i \in \{0,1\}$. We use ε to represent the empty string. Equivalently, binary strings can be thought of as ordered tuples $(a_1, \dots, a_n)$ for $a_i \in \{0,1\}$, and thus an element of $\{0,1\}^* = \bigcup_{k=0}^{\infty} \{0,1\}^k$, but where we are just concatenating them instead of using a comma.

Example
Let $B = \{0,1\}^*$ be the set of all binary strings. Then by the product lemma,

$$\Phi_B^{\text{length}}(x) = \Phi_{\{0,1\}^*}^{\text{length}}(x) = \sum_{k=0}^{\infty} \Phi_{\{0,1\}^k}(x) = \sum_{k=0}^{\infty} \Phi_{\{0,1\}}^k = \frac{1}{1 - \Phi_{\{0,1\}(x)}} = \frac{1}{1-2x}$$

where $\Phi_{\{0,1\}(x)} = x^{\text{length}(0)} + x^{\text{length}(1)} = x + x = 2x$. This gives $\Phi_B^{\text{length}}(x) = \frac{1}{1-2x} = 1 + 2x + 4x^2 + 8x^3 + \dots$, which is exactly as expected.

Definition
A regular expression (on binary strings) is defined recursively as follows:
(1) - $\varepsilon, 0, 1$ are regular expressions.

(2) - If R and S are regular expressions, then so is $R \smile S$ (read as "or"), where $\smile$ represents taking the "union" $\{R, S\}$.
(3) If R and S are regular expressions, then so is the concatenation RS, and more generally R^k for $k \geq 0$ is a regular expression (k-fold concatenation).
(4) If R is a regular expression, so is $R^* = \varepsilon \smile R \smile R^2 \smile \dots$.

Example
$0, 1$ are regular expressions, so too is 00, 10, so therefore so too is $00 \smile 10$ (representing $\{00, 10\}$), hence so is $(00 \smile 10)(00 \smile 10)$, which represents the binary strings $\{0000, 0010, 1000, 1010\}$.

Example
Consider $(0(00 \smile 11)^2)^*$. This is a regular expression. But what does it mean? We see ε is a string described by this regular expression. Otherwise, the expression $0(00 \smile 11)^2$ describes the regular expressions 00000, 00011, 01100, 01111 (start with a 0, then 2 of either 00 or 11), and then we can take all possibles concatenations of these 4 strings, to get 16 total choices of length 10, 64 total choices of length 15, etc.

Often for regular expressions, we wish to count the number of binary strings represented by this expresssion of a particular length. In the above example, there is one word of length 0 (ε), 4 words of length 5, 4^2 words of length 10, and more generally 4^n words of length $5n$, and so we can create a generating series where the coefficients are the number of binary strings of length n matched by the given expression, where in this example we have $\sum_{n=0}^{\infty} 4^n x^{5n} = \frac{1}{1-4x^5}$.

Definition
Let $\mathcal{R}$ and $\mathcal{S}$ be sets of binary strings. We define $\mathcal{RS}$ to be the set of all concatenations of strings in $\mathcal{R}$ with strings in $\mathcal{S}$.

Example
Suppose $\mathcal{R}$ is all 0 strings, and $\mathcal{S} = \{1, 11, 111\}$. Then $\mathcal{RS} = \{01, 011, 0111, 011, 0011, 00111, \dots\}$, $\mathcal{SR} = \{10, 110, 1110, 100, 1100, 11100, \dots\}$, and $\mathcal{RR} = \{00, 000, 0000, \dots\} = \mathcal{R} \setminus \{0\}$, and $\mathcal{SS} = \{11, 111, 1111, 11111, 111111\}$.

Theorem
Let R and S be regular expression representing sets of words $\mathcal{R}$ and $\mathcal{S}$ respectively, with $\varepsilon, 0, 1$ regular expressions that represent the sets $\{\varepsilon\}, \{0\}, \{1\}$ respectively. Then $R \smile S$ is a regular expression representing the strings $\mathcal{R} \cup \mathcal{S}$, RS is a regular expression representing the strings $\mathcal{RS}$ and R^k is a regular expression representing the strings $\underbrace{\mathcal{R} \dots \mathcal{R}}_{k \text{ times}} = \{a_1 \dots a_k : a_i \in \mathcal{R}\}$, sometimes denoted by $\mathcal{R}^k$, and $R^* = \varepsilon \smile R \smile R^2 \smile \dots$ is a regular expression representing the set of strings $\mathcal{R}^* := \{\varepsilon\} \cup \mathcal{R} \cup \mathcal{R}^2 \cup \dots$.

Definition
Let R be a regular expression. Then $\mathcal{R}$, the set of binary strings represented by R, is called a rational language (or regular language).

Example
Not all subsets of binary strings are rational. For example, $\{0^n 1^n\}_{n=0}^{\infty} = \{\varepsilon, 01, 0011, 000111, \dots\}$ is not a rational language. The basic idea is that with a regular expression, we have no "memory", so we can't keep track of the number of 0s we began the string with with just a regular expression. In the theory of computation, rational languages correspond precisely to the things we can program without memory. Another example is $\{0^p\}_{p \text{ prime}}$ is not a rational language.

Example
Any finite set $\mathcal{R}$ is a rational language by just taking a finite number of $\smile$'s of all of the representatives

of elements of $\mathcal{R}$.

Example

Consider the two regular expressions $(1 \smile 11)^*$ and 1^*. These both give the rational language $\{\varepsilon, 1, 11, 111, \dots\}$. We see that $(1 \smile 11)^*$ represents the string 111 in three different ways: 1.1.1 (. denotes concatenation), 1.11, 11.1. On the other hand, 1^* represents the string in only one way, 1.1.1. This motivates:

Definition

We say a regular expression is unambiguous if every string in its corresponding rational language is uniquely given by a unique representation. Otherwise we say the regular expression is ambiguous.

Example

The expression $(1 \smile 0)^*(00 \smile 0)$ is ambiguous: notice that we can get $00 = 0.0$ or $\varepsilon.00$ where . denotes concatenation, where the thing on the left comes from $(1 \smile 0)^*$ and the thing on the right comes from $(00 \smile 0)$.

Example

The expression $1^*(00 \smile 0)$ is unambiguous because it must begin with a (possibly empty) substring of 1s, which can only come in 1 way from the 1^*, and ends with 0 or 00, which can only come in a unique way from $(00 \smile 0)$.

Lemma

Let R and S be regular expressions which are unambiguous, with rational languages $\mathcal{R}$ and $\mathcal{S}$. Then:
(1) $\varepsilon, 0, 1$ are all unambiguous.
(2) The regular expression $R \smile S$ is unambiguous if and only if $\mathcal{R} \cap \mathcal{S} = \emptyset$ (note the language of $R \smile S$ is $\mathcal{R} \cup \mathcal{S}$).
(3) The regular expression RS is unambiguous if and only if there is a bijection from $\mathcal{RS}$ to $\mathcal{R} \times \mathcal{S}$. That is, for every $\alpha \in \mathcal{RS}$, there is a unique $r \in \mathcal{R}$ and $s \in \mathcal{S}$ such that $\alpha = rs$.
(4) R^* is unambiguous if and only if R^k is unambiguous is unambiguous for all k and $\mathcal{R}^k \cap \mathcal{R}^n = \emptyset$ for all $k \neq n$.
Idea of Proof:
Do some fun and long definition chases. ■

Example

Consider the regular expression $(\varepsilon \smile 0)(0 \smile 00)$. Notice $(\varepsilon \smile 0)$ is unambiguous by (2) as $\{\varepsilon\} \cap \{0\} = \emptyset$, which has language $\{\varepsilon, 0\}$. Similarly, $0 \smile 00$ is unambiguous by (2), which has language $\{0, 00\}$. Now $\{\varepsilon, 0\}\{0, 00\} = \{\varepsilon.0, \varepsilon.00, 0.0, 0.00\} = \{0, 00, 000\}$, but $\{\varepsilon, 0\} \times \{0, 00\} = \{(\varepsilon, 0), (\varepsilon, 00), (0, 0), (0, 00)\}$, so we see that $(\varepsilon \smile 0)(0 \smile 00)$ is ambiguous by (3) since the Cartesian product and the concatenation are different sizes, and thus there is no bijection between them.

Theorem

Let R and S be unambiguous regular expressions with languages $\mathcal{R}$ and $\mathcal{S}$, with generating series $\Phi_{\mathcal{R}}$ and $\Phi_{\mathcal{S}}$ with respect to the weight function that takes the length of the binary strings. Then:
(1) $\varepsilon, 0, 1$ are umambiguous regular expressions with languages $\{\varepsilon\}$, $\{0\}$, $\{1\}$ with corresponding generating series $\Phi_{\{\varepsilon\}}(x) = 1, \Phi_{\{0\}}(x) = x, \Phi_{\{1\}} = x$.
(2) If $R \smile S$ is unambiguous, with language $\mathcal{R} \cup \mathcal{S}$. Then $\Phi_{\mathcal{R} \cup \mathcal{S}(x)} = \Phi_{\mathcal{R}}(x) + \Phi_{\mathcal{S}}(x)$.
(3) If RS is an unambiguous expression for $\mathcal{RS}$, then $\Phi_{\mathcal{RS}}(x) = \Phi_{\mathcal{R} \times \mathcal{S}}(x) = \Phi_{\mathcal{R}}(x)\Phi_{\mathcal{S}}(x)$.
(4) If R^* is unambiguous with language $\mathcal{R}^*$, then $\Phi_{\mathcal{R}^*}(x) = \frac{1}{1-\Phi_{\mathcal{R}}(x)}$.
Proof:
Combine the previous lemma with the sum lemma, product lemma, and geometric series respectively. ■

Example

The regular expression $0^*(100^*)^*(\varepsilon \smile 1)$ is unambiguous. Let $\mathcal{S}$ be its corresponding rational language. Let's find its generating series. 0 has generating series $\Phi_{\{0\}}(x) = x$, 0^* has generating series $\Phi_{\{0\}^*}(x) = \frac{1}{1-\Phi_{\{0\}}(x)} = \frac{1}{1-x}$. Next $\Phi_{\{10\}\{0\}^*} = \Phi_{\{1\}}(x)\Phi_{\{0\}}(x)\Phi_{\{0\}^*}(x) = \frac{x^2}{1-x}$. This gives the generating series associated to the language of $(100^*)^*$, it is $\frac{1}{1-\frac{x^2}{1-x}}$. Lastly $\Phi_{\{\varepsilon,1\}} = \Phi_{\{\varepsilon\}}(x) + \Phi_{\{1\}}(x) = 1 + x$, hence $\Phi_{\mathcal{S}}(x) = \frac{1}{1-x}\frac{1}{1-\frac{x^2}{1-x}}(1+x) = \frac{1+x}{1-x-x^2}$.

Remark
We can do this on ambiguous expressions, but the coefficents can include overcounts and be too high.

Block and Prefix Decompositions

We see that having an unambiguous regular expression allows us to construct a generating series for the language with respect to the length. One way to do this is to decompose strings in an unambiguous way, and construct the regular expression from this.

Definition For block decompositions, we will decompose a string into alternating "blocks" of 0s and 1s. A non-empty block of 0s can be represented by 00^*, and a non-empty block of 1s can be represented by 11^*. We can alternate these like $00^*11^*00^*11^*00^*$ to represent all binary string of 5 alternating blocks of 0s then 1s, and no others. We can represent all binary strings by $1^*(00^*11^*)^*0^*$ or equivalently by symmetry, $0^*(11^*00^*)^*1^*$, which are unambiguous regular expressions. This is called a block decomposition.

Example
For example, we can split 11010001110101 as 11.0.01.000.111.0.1.0.1.

Example
Let's find an unambiguous regular expression for binary strings where all blocks of 0 are even length. Notice a non-empty block of 0s of even length can be represented by $00(00)^*$. If we also wanted to allow an empty block, we can use $(00)^*$. But now we can take our regular expression for all binary strings,

$$1^*(\underbrace{00^*}_{\text{nonempty}}11^*)^*\underbrace{0^*}_{\text{possibly empty}}$$

and modify the occurrences of 0 to what we constructed above, getting that

$$1^*(\underbrace{00(00)^*}_{\text{nonempty, even length}}11^*)^*\underbrace{(00)^*}_{\text{possibly empty, even length}}$$

as an unambiguous regular expression to represent what we want.

Example
Let's find an unambiguous regular expression for binary strings where all blocks are odd length. For blocks of 1s that are possibly empty of odd length, the new expression is $\varepsilon \smile 1(11)^*$, for blocks of 0s that are nonempty of odd length, the new expression is $0(00)^*$. For blocks of 1s that are nonempty of odd length, the new expression is $1(11)^*$, and for blocks of 0s that are possibly empty of odd length, the new expression is $\varepsilon \smile 0(00)^*$. Inserting these expressions into our regular expression for all binary strings, we get

$$(\varepsilon \smile 1(11)^*)(0(00)^*1(11)^*)^*(\varepsilon \smile 0(00)^*)$$

as our regular expression for binary strings where all blocks are odd length.

Example
We can decompose 00110001001101 as 00.1.1000.100.1.10.1, where every 1 starts a part and every part

starts with 1.

Definition
In general, we can decompose a binary string into parts where every 1 starts a part and every part (except possibly the first) starts with 1. Every one of these parts looks like 10^*, and the first part which could start with 0s looks like 0^*, which gives us an unambiguous regular expression $0^*(10^*)^*$ to capture all binary strings. Symmetrically, we could do $1^*(01^*)^*$, or use suffixes instead of prefixes. This is called prefix decomposition.

Example
Let's find an unambiguous regular expression to represent strings where every 1 is followed by at least two 0s. Here 10^* is a 1 followed by any number of 0s, so we can modify this to give 1000^*, where every 1 is followed by at least two 0s. Inserting this expression into our regular expression for all binary strings, we get $0^*(1000^*)^*$ as our needed regular expression.

Example
We have an unambiguous regular expression for all words where 1 is followed by at least two 0s. Let this language be $\mathcal{R}$. Let's find $\Phi_{\mathcal{R}}(x)$. Here the regular expression is $0^*(1000^*)^*$. Notice the generating series for the part coming from 0^* is $\frac{1}{1-x}$. The generating series corresponding to 1000^* is $\frac{x^3}{1-x}$. Putting this together gives $\left(\frac{1}{1-x}\right)\left(\frac{1}{1-\frac{x^3}{1-x}}\right)$, which can be simplified to $\frac{1}{1-x-x^3}$.

Recursive Decompositions

Another way we can describe a language is recursively. It is possible that such a language is not a rational language. Despite this, we can often still use this decomposition to say something meaningful about the language via generating series. We need sduch a decomposition to be unambiguous for this to work.

Example
Let $\mathcal{S}$ be the set of binary strings where all blocks of 1s are even, and all blocks of 0s are of length divisible by 3. For any word in $\mathcal{S}$, it is ε, or it starts with 0, or it starts with 1. As all blocks of 0s are divisible by 3, if it starts with 0, it starts with 000. Similarly for 1, it would start with 11. Hence S is the expression of the form $S = \varepsilon \smile 11S \smile 000S$. This allows us to say something about the generating series.

$$\Phi_{\mathcal{S}}(x) = \Phi_{\varepsilon}(x) + \Phi_{\{11\}}(x)\Phi_{\mathcal{S}}(x) + \Phi_{\{000\}}\Phi_{\mathcal{S}}(x) = 1 + x^2\Phi_{\mathcal{S}}(x) + x^3\Phi_{\mathcal{S}}(x)$$

implying

$$\Phi_{\mathcal{S}}(1 - x^2 - x^3) = 1 \implies \Phi_{\mathcal{S}}(x) = \frac{1}{1 - x^2 - x^3}$$

Note that this language is in fact rational, and has block decomposition $(11)^*(000(000)^*11(11)^*)^*(000)^*$.

Example
Let $\mathcal{R} = \{0^n1^n\}_{n=0}^{\infty}$. As shown previously, $\mathcal{R}$ is not rational. We see $\mathcal{R}$ is either empty, or starts with a 0 and ends with a 1, and the middle part is in $\mathcal{R}$. This gives us $R = \varepsilon \smile (0R1)$, hence its generating series looks like

$$\Phi_{\mathcal{R}}(x) = \Phi_{\varepsilon}(x) + \Phi_{\{0\}}(x)\Phi_{\mathcal{R}}(x)\Phi_{\{1\}}(x) = 1 + x^2\Phi_{\mathcal{R}}(x) \implies \Phi_{\mathcal{R}}(x) = \frac{1}{1 - x^2}$$

Example
Let $\mathcal{R}$ be the language that does not contain 001. It is useful to define $\mathcal{S}$ as the language that contains 001 exactly once, at the very end. Let R and S be their expression. Then $\mathcal{R} \cap \mathcal{S} = \emptyset$. Consider $R \smile S$.

This is either ε or ends in 0 or ends in 1. If it ends in 0, and we remove the final 0, then it will still not contain 001 (i.e. it is in $\mathcal{R}$). If ends in 1, and we remove the final 1, then it will not contain 001, and hence is in $\mathcal{R}$. This gives $R \smile S = \varepsilon \smile R0 \smile R1$.

Thus $\Phi_{\mathcal{R}}(x) + \Phi_{\mathcal{S}}(x) = 1 + 2x\Phi_{\mathcal{R}}(x)$. Now consider a word $r_1r_2\ldots r_n001$ represented by S. This can't contain a 001 anywhere in $r_1r_2\ldots r_n00$ by assumption, which occurs if and only if $r_1r_2\ldots r_n$ does not contain a 001. This gives us $S = R001$. Hence $\Phi_{\mathcal{S}}(x) = x^3\Phi_{\mathcal{R}}(x)$, and so $\Phi_{\mathcal{R}}(x) = \frac{1}{1-2x+x^3}$.

Example

Let's do the same example as above, except exclude 000 instead of 001. As before, we have $\Phi_{\mathcal{R}}(x) + \Phi_{\mathcal{S}}(x) = 1 + 3x\Phi_{\mathcal{R}}(x)$. Now consider $00 \in \mathcal{R}$. We see $00.000 \notin \mathcal{S}$: the triple 0 occurs three times, two of them not at the end. If $r_1\ldots r_n = r_1\ldots r_{n-2}00 \in \mathcal{R}$, then $r_1r_2\ldots r_n000 = (r_1r_2\ldots r_{n-2}000)(00) \in \mathcal{S}(00)$, where $r_1r_2\ldots r_{n-2}000 \in \mathcal{S}$, and similarly if $r_1\ldots r_n = r_1r_2\ldots r_{n-1}0 \in \mathcal{R}$, $r_{n-1} \neq 0$, then $r_1\ldots r_{n-1}0000 = (r_1r_2\ldots r_{n-1}000)(0) \in \mathcal{S}0$. This gives $S \smile S0 \smile S00 = R000$, so $\Phi_{\mathcal{S}}(x)(1+x+X^2) = x^3\Phi_{\mathcal{R}}(x)$, and the rest is algebra.

Theorem

Let $k \in \{0,1\}^*$ be a nonempty binary string of length n. Let $\mathcal{C}$ be the collection of all nonempty proper suffixes γ of k such that there exists a nonempty prefix η of k with $\eta k = k\gamma$. Let $C(x)$ be the generating series of $\mathcal{C}$ with respect to length. Then the generating series for the language of binary strings not containing k is

$$\Phi(x) = \frac{1 + C(x)}{(1-2x)(1+C(x)) + x^n}$$

Proof:

In the original method we always have $R \smile S = \varepsilon \smile R0 \smile R1$, where R is the regular expression avoiding k with language $\mathcal{R}$, and S is the regular expression matching only k at the end with language $\mathcal{S}$. The second relationship depends on $\mathcal{C}$. Let $\mathcal{C} = \{\gamma_1, \ldots, \gamma_m\}$. We get a new relation $Rk = S \smile S\gamma_1 \smile \cdots \smile S\gamma_m$. From these two relations we get the generating series $\Phi_{\mathcal{R}}(x) + \Phi_{\mathcal{S}}(x) = 1 + 2x\Phi_{\mathcal{R}}(x)$, and $x^n\Phi_{\mathcal{R}}(x) = \Phi_{\mathcal{S}}(x) + \sum_{\gamma_i \in \mathcal{C}} x^{\ell(\gamma_i)}\Phi_{\mathcal{S}}(x) = 1 + C(x)\Phi_{\mathcal{S}}(x)$, so $\Phi_{\mathcal{S}}(x) = \frac{x^n}{1+C(x)}$, and doing the algebra from here we get the result. ■

Example

Let $k = 000$. Notice the suffixes of k are 00 and 0. In this case $0.000 = 000.0$, and $00.000 = 000.00$ so in this case $\mathcal{C} = \{0, 00\}$, and thus $C(x) = x + x^2$, giving

$$\Phi(x) = \frac{1 + x + x^2}{(1-2x)(1+x+x^2) + x^3} = \frac{1 + x + x^2}{1 - x - x^2 - x^3}$$

Example

Let $k = 1010$, $\mathcal{R}$ the language avoiding k, and $\mathcal{S}$ the language where k occurs exactly once at the end. We have $R \smile S = \varepsilon \smile R0 \smile R1$. What happens if we append k to the end of a word in $\mathcal{R}$? If the word in R ended with 10, then $r_1\ldots r_m10k = r_1\ldots r_m101010$, which is in $S10$. In this case, the 10 corresponds to a $\gamma \in \mathcal{C}$ as $10k = k10 = 101010$. If the word does not end in 10, then we are fine, and $r_1\ldots r_n1010 \in \mathcal{S}$ (notice the nonempty suffixes of $k = 1010$ are $010, 10, 0$, where $10100 = k0 \neq \eta k$ for all η since it does not end in 10, and $k010 = 1010010$ also cannot end with k). From here we get $Rk = S \smile S10$.

Recurrences

Example

Let $g_0 = 1, g_1 = 1$, and $g_n = 2g_{n-1} + g_{n-2}$ for all $n \geq 2$. What can we say about $G(x) = \sum_{n=0}^{\infty} g_nx^n$? Can we find a closed form for it? What can we do with this closed form? Let's first find a closed form for $G(x)$. Then

$$G(x) = \sum_{n=0}^{\infty} g_n x^n = 1 + x + \sum_{n=2} x^n = 1 + x + \sum_{n=2}^{\infty}(2g_{n-1} + g_{n-2})x^n = 1 + x + \sum_{n=2}^{\infty} 2g_{n-1}x^n + \sum_{n=2}^{\infty} g_{n-2}x^n$$

$$= 1 + x + \sum_{n=2}^{\infty} 2g_{n-1} + x^2 \sum_{n=0}^{\infty} g_n x^n = 1 + x + \sum_{n=2}^{\infty} 2g_{n-1} + x^2 G(x) = 1 + x + 2x \sum_{n=1}^{\infty} g_{n-1}x^{n-1}$$

$$= 1 + x + 2x(-1 + G(x)) + x^2 G(x) \implies G(x) = \frac{1-x}{1-2x-x^2}$$

Now, consider the denominator $1 - 2x - x^2$ of $G(x)$. We factor it as $1 - 2x - x^2 = (1 - \alpha x)(1 - \beta x)$, for reasons that will become clear later. Writing $x = 1/y$ and multiplying by y^2, we want to find the roots of $y^2 - 2y - 1$, which by the quadratic formula are $\frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}$. This gives $y^2 - 2y - 1 = (y - (1 + \sqrt{2}))(y - (1 - \sqrt{2}))$, which equivalently gives $1 - 2x - x^2 = (1 - (1 + \sqrt{2})x)(1 - (1 - \sqrt{2})x)$. This gives us

$$G(x) = \frac{1-x}{(1 - (1+\sqrt{2})x)(1 - (1-\sqrt{2})x)}$$

We can now do a partial fraction decomposition, to get A, B such that

$$\frac{1-x}{(1 - (1+\sqrt{2})x)(1 - (1-\sqrt{2})x)} = \frac{A}{1 - (1+\sqrt{2})x} + \frac{B}{1 - (1-\sqrt{2})x}$$

which equivalently gives

$$1 - x = B(1 - (1 - \sqrt{2}x)) + B(1 - (1 + \sqrt{2}x))$$

With a bit of inspection, we see $A = \frac{1}{2}$, $B = \frac{1}{2}$ works. This gives us

$$G(x) = \sum_{n=0}^{\infty} g_n x^n = \frac{1-x}{1-2x-x^2} = \frac{(1/2)}{1 - (1+\sqrt{2})x} + \frac{(1/2)}{1 - (1-\sqrt{2})x} = \frac{1}{2}\sum_{n=0}^{\infty}(1+\sqrt{2})^n x^n + \frac{1}{2}\sum_{n=0}^{\infty}(1-\sqrt{2})^n x^n$$

$$= \sum_{n=0}^{\infty} \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2} x^n$$

and so $g_n = \dfrac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2}$. That is, we have found a closed form for the coefficients of our generating series! Moreover, we notice that in this case, that $1 - \sqrt{2} \approx -0.44$, and so $(1 - \sqrt{2})^n \to 0$ as $n \to \infty$, so for large n, $g_n \approx \frac{1}{2}(1 + \sqrt{2})^n$. Furthermore, since all g_n are integers, we see $\frac{1}{2}(1 + \sqrt{2})^n$ is very close to an integer for large n, and that in general, $\frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2}$ is an integer for all $n \in \mathbb{N}$ (since g_n is), which is not intuitively obvious.

Definition

Let $a_1, \ldots, a_d \in \mathbb{C}$ (though we typically work with integers). Let $\{g_i\}_{i=0}^{\infty}$ be a sequence with the first m terms (called the initial conditions) $g_0, g_1, \ldots, g_m$ specified, with $m \geq d - 1$. Then the sequence is a homogeneous linear recurrence relation if for all $n \geq m + 1$, we have $g_n + a_1 g_{n-1} + \cdots + a_d g_{n-d} = 0$.

Example

Our previous example is an example of such a relation. Here, $g_1 = g_0 = 1$, and $g_n - 2g_{n-1} - g_{n-2} = 0$ for all $n \geq 2$. Here $a_1 = -2, a_2 = -1$. Now notice in the previous example we had $G(x) = \frac{p(x)}{1 + a_1 x + a_2 x^2}$ for some polynomial $p(x)$. It's not a coincidence a_1, a_2 showed up in the denominator.

Theorem
Let $g_n + a_1 g_{n-1} + \cdots + a_d g_{n-d} = 0$ be a linear homogeneous recurrence relation. Then

$$\sum_{n=0}^{\infty} g_n x^n = \frac{b_m x^m + \cdots + b_0}{1 + a_1 x + a_2 x^2 + \cdots + a_d x^d}$$

where $b_k = g_k + a_1 g_{k-1} + \cdots + a_d g_{k-d}$ (taking $g_n = 0$ if $n < 0$) for $0 \leq k \leq m$ with m the number of initial conditions.
Proof:
In the course notes, Alex will maybe transcribe later. ■

Example
Let $G(x) = \sum_{n=0}^{\infty} g_n x^n = \frac{2x^2-4x+3}{(1-2x)^2(1+x)}$. Let's find a closed form for g_n, find initial conditions for a recurrence relation, and then find the recurrence relation. By partial fraction decomposition, there exist a, b, c such that

$$\frac{2x^2 - 4x + 3}{(1-2x)^2(1+x)} = \frac{a}{1-2x} + \frac{b}{(1-2x)^2} + \frac{c}{(1+x)}$$

and so

$$2x^2 - 4x + 3 = a(1-2x)(1+x) + b(1+x) + c(1-2x)^2$$

Evaluating at $x = 0$ tells us $c = 1$, evaluating at $x = \frac{1}{2}$ tells us $b = 1$, which substituting these in, tells us $a = 1$. Therefore,

$$\frac{2x^2 - 4x + 3}{(1-2x)^2(1+x)} = \frac{1}{1-2x} + \frac{1}{(1-2x)^2} + \frac{1}{1+x}$$

This gives

$$G(x) = \sum_{n=0}^{\infty} 2^n x^n + \frac{1}{(1-2x)^2} + \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} (2^n + \binom{n+1}{1} 2^n + (-1)^n) x^n$$

$$= \sum_{n=0}^{\infty} (2^{n+1} + 2^n + (-1)^n) x^n$$

This allows us to read off the initial conditions $g_0 = 3, g_1 = 5, g_2 = 17$, and the original denominator gives us the recurrence relation: we have $(1-2x)^2(1+x) = 1-3x+4x^3$, which gives $g_n - 3g_{n-1} + 0g_{n-2} + 4g_{n-3} = 0$, or in other words, $g_n = 3g_{n-1} - 4g_{n-3}$.

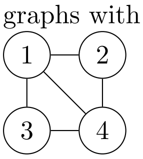
We've seen that given a linear homogeneous recurrence relation, there are several ways we can describe them:
(1) As a set of initial conditions $g_0, g_1, \ldots, g_{d-1}$, with a recurrence relation $g_n + a_1 g_{n-1} + \cdots + a_d g_{n-d} = 0$ for all $n \geq d$.
(2) As a generating series $G(x) = \sum_{n=0}^{\infty} g_n x^n = \dfrac{P(x)}{Q(x)}$.
The theorem above gives the connection between (1) and (2): $Q(x) = 1 + a_1 x + \cdots + a_d x^d$.
(3) Take $Q(x)$ as above, and assume $Q(x) = (1 - \lambda_1 x)^{\alpha_1} \ldots (1 - \lambda_k x)^{\alpha_k}$. Then there exist polynomials $p_1(x), \ldots, p_k(x)$ with $\deg p_i(x) \leq \alpha_i - 1$ such that $g_n = p_1(n)\lambda_1^n + \cdots + p_k(n)\lambda_k^n$.

To go from (3) to (2), given a polynomial $g_n = p_1(n)\lambda_1^n + \cdots + p_k(n)\lambda_k^n$, take $Q(x) = (1 - \lambda_1 x)^{\deg p_1 + 1} \ldots (1 - \lambda_k x)^{\deg p_k + 1}$ (see Theorem 4.14 and Exercise 4.11), and take $P(x)$ by using g_n to get the necessary number of initial conditions (the degree of $Q(x)$), and take $b_k = g_k + a_1 g_{k-1} + \cdots + a_k g_0$ with $P(x) = b_0 + b_1 x + \cdots + b_k x^k$.
To go from (2) to (3), just look at the term-by-term coefficients of the power series as functions of n.

Graph Theory

Definition
A graph G is a collection of vertices, denoted $V(G)$, say $\{v_1, \ldots, v_n\}$ and a collection of unordered pairs (edges) between these vertices, $E(G) = \{(v_{i_1}, v_{j_1}), (v_{i_2}, v_{j_2}), \ldots\}$. For now, we assume the number of vertices is finite. We will assume that there is at most one edge between two vertices, and that the edges are undirected (i.e. pairs in the edge set are unordered).

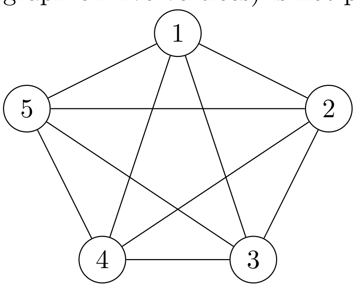
Example
Let G be a graph with $V(G) = \{1, 2, 3, 4\}$ and $E(G) = \{(1, 2), (1, 3), (1, 4), (2, 4), (3, 4)\}$. We often draw graphs with circles:



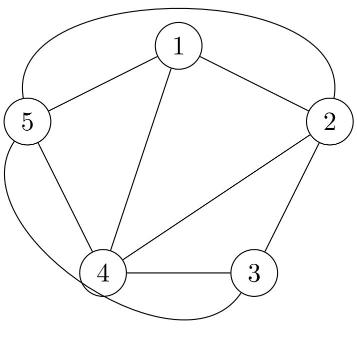
Definition
We say two vertices are adjacent if there is an edge between them, for example, in the above graph, 1 and 2 are adjacent, 2 and 3 are not. We define the neighbours of a vertex to be all the vertices that it is adjacent to. In the above graph, the neighbours of 2 are 1 and 4.

Definition
If we can draw the graph in such a way that no edges cross each other, then the graph is called planar. The previous example is planar.

Example
The graph G on five vertices where there is an edge between every pair of vertices (the so-called complete graph on five vertices) is not planar.



In this example, removing any edge makes the graph planar (by rearranging how we draw the edges). For instance, removing $(1, 3)$,



Example

There are a huge number of applications of graph theory, such as
(1) Networks
(2) Travelling salesperson problem
(3) Colouring countries on a map.
There are a lot of variations of graphs which are interesting:
(1) Allowing multiple edges in pairs
(2) Loops in graphs
(3) Adding directions to edges
(4) Attaching weights to edges
(5) Allow infinite graphs.
(6) Take edges as unordered tuples instead of unordered pairs.

Definition
Define the degree of a vertex v, $\deg(v)$, to be the number of edges touching the vertex. For instance, in the above example, (3) has degree 3. It is easy to see $\deg(v)$ is the number of neighbours of v.

Theorem
Let G be a graph.

$$\sum_{v\in V(G)} \deg(v) = 2|E(G)|$$

Proof
We see each edge is connected to two vertices, and thus adds 1 to each of their degrees. Summing over all edges gives the result. ■

Corollary:
Let G be a graph. The number of vertices $v \in V(G)$ with odd degree is even.
Proof:
By the above theorem, write

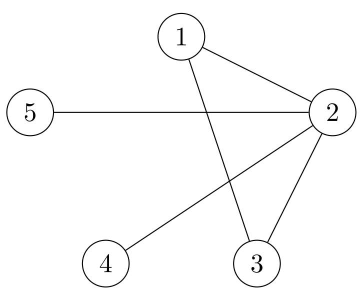
$$2|E(G)| = \sum_{v\in V(G)} \deg(v) = \sum_{v\in V(G),\ \deg v \text{ odd}} \deg(v) + \sum_{v\in V(G),\ \deg v \text{ even}} \deg(v)$$

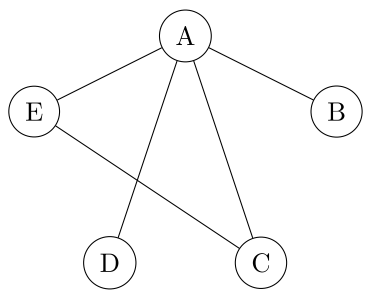
and thus the sum of odd degrees is equal to an even number $2|E(G)|$ minus an even number $\sum_{v\in V(G),\ \deg v \text{ even}} \deg(v)$, and so must be even, and thus we must have an even number of odd degrees. ■

Corollary
Let G be a graph. The average degree of a vertex is $\frac{2|E(G)|}{|V(G)|}$. ■

Definition
Let G_1 and G_2 be graphs. We say G_1 is isomorphic to G_2 if there is a bijection $f : V(G_1) \to V(G_2)$ with the additional property that $(v_1, v_2) \in E(G_1)$ if and only if $(f(v_1), f(v_2)) \in E(G_2)$.

Example

The graphs 

and 

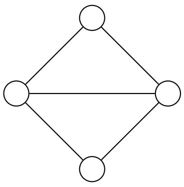
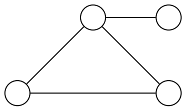
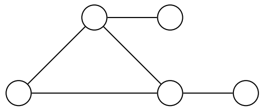
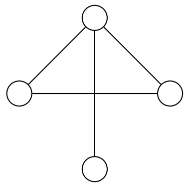
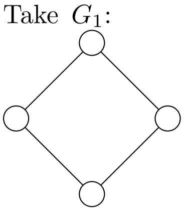
are isomorphic. To see

this, consider the bijection $1 \mapsto C$, $5 \mapsto D$, $4 \mapsto B$, $3 \mapsto E$, $2 \mapsto A$, where it is easy to see this is a bijection from $V(G_1)$ to $V(G_2)$. We can check the edges: $(1,2) \mapsto (E,A)$, $(1,3) \mapsto (E,C)$, $(2,3) \mapsto (A,C)$, $(2,5) \mapsto (A,D), (2,4) \mapsto (A,B)$, which are all edges in the second graph.

Facts

Let G_1 and G_2 be isomorphic graphs. Then they have the same number of vertices, the same number of edges, and the degree of the vertices are preserved under the isomorphism.

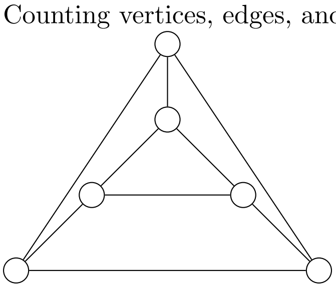
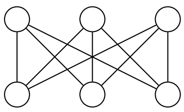
Example

Take G_1:  G_2 :  G_3 :  G_4 :  G_5: 

We see G_1 has 5 edges, 4 vertices, 0 vertices of degree 1, G_2 has 4 edges, 4 vertices, 1 vertex of degree 1, G_3 has 5 edges, 5 vertices, 2 vertices of degree 1, G_4 has 4 edges, 4 vertices, 1 vertex of degree 1, and G_5 has 4 edges, 4 vertices, and 0 vertices of degree 1. Thus the only possible isomorphisms in this list is G_2 with G_4, and indeed they are isomorphic (send the one degree 1 vertex to each other, the rest do the "identity" map).

Example

Counting vertices, edges, and degrees is only necessary not sufficient:

 and 

have the same number of vertices, edges, and degrees, but are not isomorphic (the left contains a cycle of length 3, the right does not).

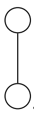
Definition

We say a graph G is k-regular if $\deg(v) = k$ for all $v \in V(G)$. We say a graph is regular if it is k-regular for some k.

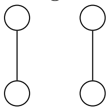
Example

The graph with n vertices and no edges is 0-regular.

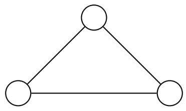
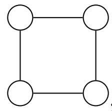
Example

The only 1-regular graph is the graph with 2 connected vertices: . As the number of vertices of odd degree is even, and 1 is odd, we have that 1-regular graphs have an even number of vertices. In general a

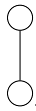
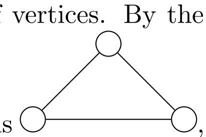
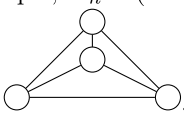
1-regular graph looks like n disconnected copies of the 1-regular graph with 2 vertices, such as $(n = 2)$:



Example

Some 2-regular graphs are the "strict" cycles like  and , and arbitrary disconnected copies of these.

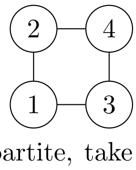
Example
In general, the smallest k-regular graph is the graph of $k + 1$ vertices $(v_1, \ldots, v_{k+1})$ where you put an edge between every pair of vertices.

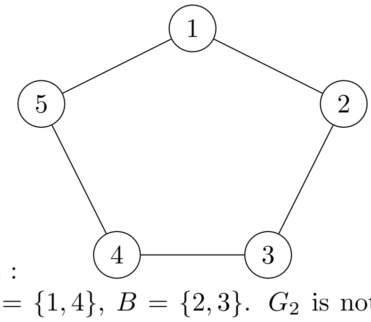
Definition
A complete graph K_n is a graph with n-vertices, and an edge between every pair of vertices. By the above example, K_n is $(n-1)$-regular. For example, K_1 is one vertex, K_2 is , and K_3 is , and K_4 is .

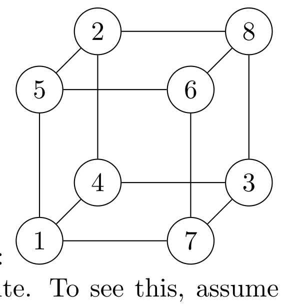
Proposition
K_n has $\frac{n(n-1)}{2}$ edges.
Proof:
$2|E(K_n)| = \sum_{v \in V(K_n)} \deg v = n(n-1)$ ■

Bipartite graphs

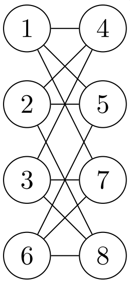
Definition
We say a graph G is bipartite if we can divide the vertices $V(G)$ into two disjoint sets A and B such that all edges have one end in A and one end in B.
Example

Take G_1: 

G_2 : 

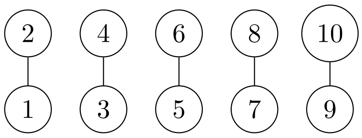
and G_3: 

G_1 is bipartite, take $A = \{1, 4\}$, $B = \{2, 3\}$. G_2 is not bipartite. To see this, assume for the sake of contradiction it is. Without loss of generality assume $1 \in A$. As there is an edge from 1 to 5 and 1 to 2, as there are no edges from A to A, we have $2, 5 \in B$. There is an edge from 5 to 4, hence $4 \in A$. Similarly there is an between 2 and 3, hence $3 \in A$. But there is an edge from 3 to 4, which is an edge from a vertex in A to a vertex in A, so G_2 cannot be bipartite. The argument works to show G_3 is bipartite, say $1 \in A$. Then we must have $4, 7, 5 \in B$, and so $6, 3, 2 \in A$, and so $8 \in B$. We can check this gives a valid partition, so G_3 is bipartite. We can see this more clearly if we rearrange the vertices (but don't change the edges) to



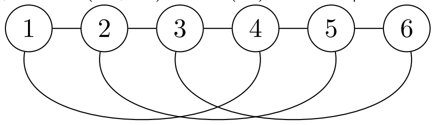
from which it is clear we have a bipartite graph, this is a standard way to draw them.

Example

Let G be a 1-regular graph on 10 vertices. 

This is bipartite, for instance $A = \{1, 3, 5, 7, 9\}, B = \{2, 4, 6, 8, 10\}$. However there are multiple partitions other than this one, for instance $A = \{1, 4, 5, 8, 9\}$ and $B = \{2, 3, 6, 7, 10\}$.

Example

Let G be a graph on 100 vertices, $V(G) = \{1, \ldots, 100\}$, with $(v_1, v_2) \in E(G)$ when $|v_1 - v_2| = 1$ or $|v_1 - v_2| = 3$. The first couple vertices of this graph are 

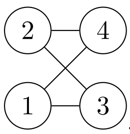
(where 4 is connected to 7, 5 is connected to 8, etc.)

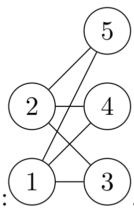
We see that the odd numbers do not have edges between each other, and even numbers do not have edges between each other, so we can take $A = \{1, 3, 5, \ldots, 99\}$ $B = \{2, 4, 6, \ldots, 100\}$, so this graph is bipartite.

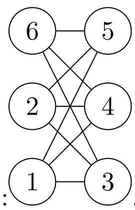
Definition

We say G is a complete bipartite graph if it is bipartite and given any (equivalently some) partition A and B of $V(G)$, every single vertex in A has an edge to every vertex in B. This is typically denoted $K_{n,m}$ where $|A| = n, |B| = m$. Since A and B are disjoint, $|V(K_{n,m})| = n + m$, and since every element of A has an edge to an element of B, and no edges between B, we get m edges for each of the n vertices in A, so $|E(K_{n,m})| = nm$.

Example

$K_{2,2}$: 

, $K_{2,3}$: 

, $K_{3,3}$: 

.

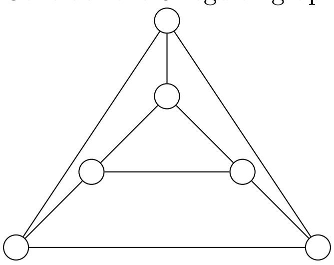
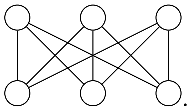
Theorem

Let G_1 be isomorphic to G_2. Then G_1 is bipartite if and only if G_2 is bipartite.

Proof:

Let $f : V(G_1) \xrightarrow{\cong} V(G_2)$ be said isomorphism, and suppose G_1 is bipartite, with partitioning sets A and B. Then $f(A)$ and $f(B)$ partition G_2 because f preserves edge relations. ■

Example
Consider the 3-regular graphs on 6 vertices we saw before,


and

.

The right graph is bipartite, the left is not (it contains an odd cycle), so these graphs are not isomorphic.

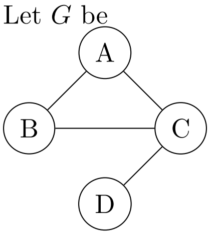
Example
Let $K_{n,m}$ be a k-regular bipartite graph for $k \neq 0$. We claim $n = m$. Indeed, if $K_{n,m}$ is k-regular, we see every vertex in A has degree k, so there are nk edges total in $K_{n,m}$. Similarly, the same argument applied to B shows there are km edges total, so either $k = 0$ or $m = n$.

Specifying Graphs

How do we specify a graph?
-We can draw a picture, which works well if the graph is small, but is completely unhelpful for large sizes. Such a picture is helpful to figure out certain properties, like planarity and bipartiteness.
-We can also explicitly specify the vertices and edges. This is quite effective when the number of edges is small, but hard to determine certain properties from it like planarity.
-We can give the vertices and a rule for the edges.
For example, let $V(G)$ be the set of all subsets of $\{1, 2, 3\}$. We say $(A, B) \in E(G)$ if $A \cap B = \emptyset$, where the vertex $\emptyset$ has the largest degree, and $\{1, 2, 3\}$ has the smallest degree.
-We can use an adjacency matrix:

Definition
Let G be a graph with $V(G) = \{v_1, \ldots, v_n\}$. The adjacency matrix of G is the $n \times n$ matrix A, where $A_{ij} = 1$ if $(v_i, v_j) \in E(G)$ and is 0 otherwise (sometimes the entries are indexed by $V(G)$).

Example
Let G be



Then its adjacency matrix is

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

where we index by A, B, C, D (in that order) for the rows and columns.

Remark
Observe that because there are no loops in our graph (no edges from a vertex to itself), all diagonal entries of an adjacency matrix are 0, and because our graph is undirected, the adjacency matrix is symmetric. Because we do not allow multiple edges, all entries are 0 or 1. We will see later that powers of the adjacency matrix give us information about walks on the graph. Moreover, we see that adjacency matrices determine our graph, though this is not a bijective correspondence (we can permute the ordering of the vertices).

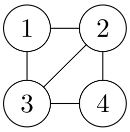
Definition
We can also specify an adjacency list, which for the above example is:

vertex	neighbours
A	BC
B	AC
C	ABD
D	C

Paths and Cycles

Definition
Let G be a graph. We define a walk from $v_0 \in V(G)$ to $v_n \in V(G)$ as a sequence of vertices $v_0, \dots, v_n$ such that $(v_i, v_{i+1}) \in E(G)$ for $i = 0, \dots, n-1$. We call n the length of the walk (as there are n edges), and note that we make no other restrictions on the vertices (in particular, repeats are allowed).

Example
Let's find all walks of length 2 starting at 1 of



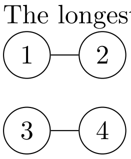
There are 6 such walks, $1-3-2, 1-2-3, 1-2-1, 1-3-1, 1-2-4, 1-3-4$.

Definition
Let G be a graph. A path on G is a walk where all vertices are distinct.

Example
Given the above graph, it has 4 paths of length 2 starting at 1, $1-3-2, 1-2-3, 1-2-4, 1-3-4$.

Fact
The longest path in a graph G has length at most $|V(G)|-1$, because all vertices must be distinct. This is only an upper bound:

Example
The longest path in

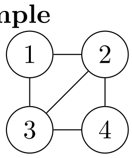


is of length 1.

Theorem
Let G be a graph, with $x, y \in V(G)$. If there exists a walk from x to y, then there exists a path from x

to y.
Proof:
If $x = y$ we are done, so suppose $x \neq y$. Let $v_0 = x, v_1, \ldots, v_{n-1}, v_n = y$ be a walk from x to y. If all vertices are distinct, we are done. Thus assume $v_i = v_j$ for some $0 \leq i < j \leq n-1$. Consider the walk $v_0, v_1, \ldots, v_i, v_{j+1}, \ldots, v_n$ (deleting the subwalk between v_i and v_j). Because $v_i = v_j$, there is still an edge between v_i and v_{j+1}, so this is a walk from $v_0 = x$ to $v_n = y$. Repeat this process until all vertices are distinct (it will terminate since we have a walk of finite length, and we are shortening it), which in the end gives us a path from x to y. ■

Example

Consider 

A walk from 1 to 4 is $1, 2, (3), 2, 4, (3), 2, 1, 3, 4$. We can delete the subwalk between the paranthesized vertices to get $(1), 2, 3, 2, (1), 3, 4$, delete the paranthesized, and get $1, 3, 4$ as our path. Alternatively we could do $1, 2, (3), 2, 4, 3, 2, 1, (3), 4$ which gives $1, 2, 3, 4$ as our path. Thus this algorithm always gives a path, though not always the same one (nor of a specific length!).

Theorem

Let G be a graph with $x, y, z \in V(G)$. If there is a path from x to y and from y to z, then there is a path from x to z.
Proof:
Concatenate the path from x to y and the path from y to z to get a walk from x to z, and apply the previous result. ■

Definition

A graph G is connected if for any $v_1, v_2 \in V(G)$, there exists a path from v_1 to v_2.

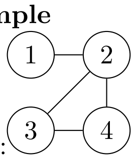
Remark

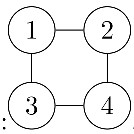
We could use a walk in the definition instead of a path by the above theorem.

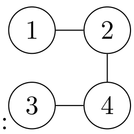
Definition

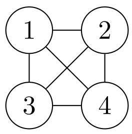
Let G_1 and G_2 be graphs. We say G_1 is a subgraph of G_1 if $V(G_1) \subseteq V(G_2)$ and $E(G_1) \subseteq E(G_2)$.

Example

Take G_1 : 

G_2 : 

, G_3 : 

G_4 : 

All graphs are subgraphs of themselves, all graphs are subgraphs of G_4 (it is complete), G_3 is a subgroup of G_1 and G_2.

Facts

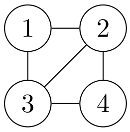
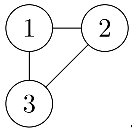
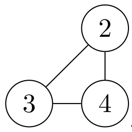
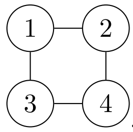
-For G_1 a subgraph of G_2, we have $|E(G_1)| \leq |E(G_2)|, |V(G_1)| \leq |V(G_2)|$.
-If G_2 is bipartite, provided G_1 does not only contain vertices from one partition, then G_1 is also bipartite.
-Every graph G_1 is a subgraph of the complete graph on vertices $V(G_1)$.

Definition

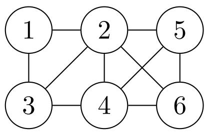
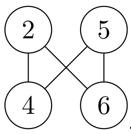
If G_1 is a subgraph of G_2 and $V(G_1) = V(G_2)$ then we say G_1 is a spanning subgraph of G_2. If in addition $E(G_1) \neq E(G_2)$, then G_1 is called a proper spanning subgraph.

Fact
A spanning subgraph of a bipartite graph is bipartite.

Definition
A connected 2-regular subgraph G_1 of a graph G_2 is called a cycle.

Example
The cycles of  are  , , and .

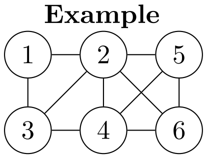
Theorem
Let G be a graph such that $\deg(v) \geq 2$ for all $v \in V(G)$. Then G contains a cycle.
Proof:
Because $\deg(v) \geq 2$ for all $v \in V(G)$, there exists a path $v_1 - v_2 \cdots - v_n$ of maximal length in G. Since $\deg(v_n) \geq 2$, it is adjacent to some vertex that is not v_{n-1}, say $x \in V(G)$ such that $(v_n, x) \in E(G)$ with $x \neq v_{n-1}$. If $x \neq v_i$ for all i in our path, then this contradicts maximality of the path, so we must have $x = v_i$ for some $i \in \{1, \ldots, n-1\}$, giving that the subgraph with vertices $\{v_i, \ldots, v_n\}$ (and only the edges determined by the path) is a cycle (it is clearly connected), and it is 2-regular because it only contains the edges of the path. ■

Example
Let's find a cycle in G :  A longest path is $1--3--2--4--5--6$. 6 is connected to 2 as well, so connected 2 to 6 and remove everything from the path before 2, getting $2--4--5--6(--2)$, giving the cycle .

Definition
We say the girth of a graph G is the size of the smallest cycle, denoted by $g(G)$.

Definition
We say a cycle is a Hamiltonian cycle if it contains every vertex.

Definition
We say a path is a Hamiltonian path if it visits every vertex.

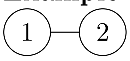
Example
 has girth $g(G) = 3$ (this is the minimal girth, and we see it is achieved by $\{1, 2, 3\}$, and we see it is a Hamiltonian path $1 - -2 - -5 - -6 - -4 - -3$ with Hamiltonian cycle generated by this path (connect 3 to 1).

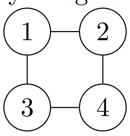
Exercise

Let K_n be a complete graph on n vertices. How many cycles of size k does K_n contain?

Fact
Let G_1 be isomorphic to G_2. Then
(1) $g(G_1) = g(G_2)$
(2) The subgraphs of G_1 correspond to subgraphs of G_2 under the isomorphism, and in particular the cycles and paths of G_1 correspond to cycles and paths of G_2, in particular if they are Hamiltonian or not.

Connected Graphs

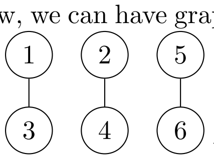
Example


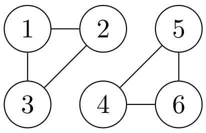
is the only 1-regular connected graph, the only 2-regular connected graphs are the cycles, for instance the square 

A complete graph is always connected.

Theorem
Let G be a graph. Let $v \in V(G)$ be such that there is a walk from v to w for all $w \in V(G)$. Then G is connected.
Proof:
Let $v_1, v_2 \in V(G)$. Then walk from v_1 to v, then v to v_2, and get the corresponding path out of it. This gives a path from v_1 to v_2, so G is connected. ■

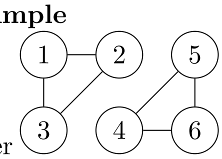
Theorem
Let G_1 be isomorphic to G_2. Then G_1 is connected if and only if G_2 is connected. ■

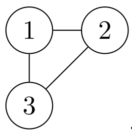
Now, we can have graphs that are disconnected, but one is more disconnected than the others. For instance 

versus 

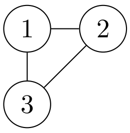
One has 2 connected components (maximal connected subgraphs), the other has 3. So we can use these to measure how disconnected a graph is.

Definition
A component of a graph G is a connected subgraph H of G such that if a subgraph H' of G contains H as a subgraph, then H' is disconnected.

Example
Consider 

Then $(1)-(2), (4)-(5)$ is not a component as it is not connected, and $(1)-(2)$ is not a component because it is a proper subgraph of the connected subgraph 

. On the other

hand,  is a component, because it is connected, and we see that it is only a proper subgraph of disconnected graphs (adding any combination of the vertices 4, 5, or 6 from the original graph make it disconnected because those are not connected to any of 1, 2, or 3).

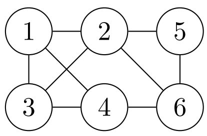
Theorem

Let G_1 be isomorphic to G_2. Then G_1 has the same number of components as G_2. ■

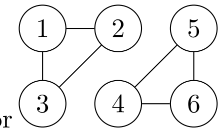
Definition

Let $X \subseteq V(G)$. We define a cut induced by X as the set of edges with one end in X and one end outside of X.

Example

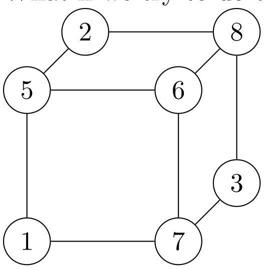
Let $X = \{1, 2, 5\}$ for the graph  Then the cut induced by X is the set of edges $(1, 3), (1, 4), (2, 3), (2, 6), (5, 6)$.

Example

Let's find a nonempty proper subset $X \subseteq V(G)$ such that the cut induced by X is empty for 
Notice that the cuts induced by $\{1, 2, 3\}$ and $\{4, 5, 6\}$ are empty.

Example

What if we try to do the above example on



Well then X must be nonempty, so without loss of generality say $1 \in X$. Then because there is an edge from 1 to 7, we must have $7 \in X$ (otherwise the cut isn't empty), and because there is an edge from 7 to 6 and 7 to 3, we must have $6, 3 \in X$, and repeating this argument, because the graph is connected, we get that $V(G) \subseteq X$, contradicting the fact that X must be proper. This generalizes:

Theorem

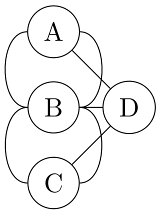
A graph G is disconnected if and only if there exists a nonempty proper $X \subsetneq V(G)$ such that the cut induced by X is empty.

Proof:

In the forward direction, assume G is disconnected. Hence there exists $x, y \in V(G)$ such that there is no path between x and y. Let X be the set of vertices connected to x by a path (the component of x). We see $x \in X$, so X is nonempty, and X is proper because $y \notin x$. Consider (v_1, v_2) in the cut induced by X. Assume $v_1 \in X, v_2 \notin X$. There is a path from x to v_1, and a path from v_1 to v_2, hence a path from x to v_2, a contradiction. Therefore, the cut induced by X is empty. Conversely, assume there exists a nonempty proper X such that the cut induced by X is empty. We want to show there is no path between some $x, y \in G$. Assume for the sake of contradiction that $x - v_1 - \cdots - v_n - y$ is such a path. Let v_k be maximal such that $v_k \in X, v_{k+1} \notin X$ (with $v_0 = x, v_{n+1} = y$). Then (v_k, v_{k+1}) is an edge in the cut induced by X, a contradiction. ■

Eulerian Circuits

Example (Bridges of Königsberg, Euler, 18th Century)
Consider the (multi)graph

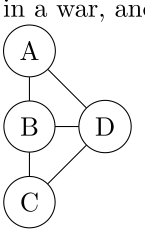


Each edge represents bridges on an island. Can we take a walk from an island, crossing each bridge exactly once, and end up at the starting island? We can translate this question to walks on our graph: Does there exist a graph starting and ending at the same vertex that goes through every edge exactly once? This is not possible for this graph. We see every time such a walk goes through a vertex, it must enter and exit via different edges. Hence if a vertex is visited k times, it has degree $2k$, i.e. has even degree, but several nodes on the graph do not have even degree, so this is impossible. Hence such a walk does not exist for this graph.

Definition
An Eulerian circuit is a walk that starts and ends at the same vertex, and goes through every edge exactly once.

Definition
An Eulerian walk is a walk that goes through every edge exactly once (and may start or end at different locations).

Example
Returning to our old example, the city of Königsberg, now called, Kaliningrad, has had bridges destroyed in a war, and so the graph of its islands and bridges now looks like



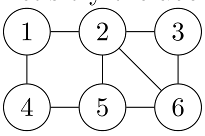
Then $B - A - D - B - C - D$ is an Eulerian walk (there is no circuit).

Is it always possible to find an Eulerian circuit if all vertices have even degree? If G is connected, then yes:

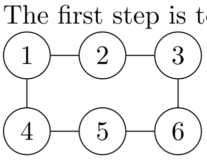
Theorem
Let G be a connected graph. G has an Eulerian circuit if and only if every vertex has even degree.
Remark: This proof works if the graph has multiedges or loops.
Proof:
In the forward direction, an Eulerian circuit contributes 2 to the degree of a vertex for each visit (entering and leaving), and since it visits each vertex exactly once, this degree must therefore be even. For the converse, we do this by induction on the size of the edge set of G. As a base case, 0 edges is trivially

true, and 1 edge is only possible if we have a loop from our one vertex to itself, which is such a circuit. If we have 2 edges, then we have $(1)-(2),(2)-(1)$ (if we allow multigraphs), for which the statement is clear, and we have 3 edges, then we have a triangle, again for which the statement is true. Now, assume the statement is true for every connected graph with m-edges or less. Notice every vertex has degree at least 2 (since even degree and connected). Hence G will have a cycle $v_1 - \cdots - v_n$. Delete this cycle from the edge set of the graph, and delete disconnected vertices as needed. Then every vertex still has even degree (every vertex has either 0 or 2 edges in the cycle), and every component has m or fewer edges, so every component has an Eulerian circuit. Furthermore, every component shares a vertex with the cycle, otherwise the graph was originally disconnected. We now glue things together, that is, take our cycle $v_1 - \cdots - v_m$, but when we hit a vertex belonging to one of the components with an Eulerian circuit, add this in as a detour, which is an Eulerian circuit. ■

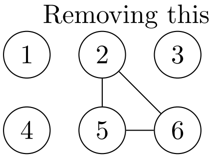
Example
Let's try the above construction on the graph



The first step is to find a cycle, for instance $1-2-3-6-5-4-1$



Removing this cycle from the graph, getting

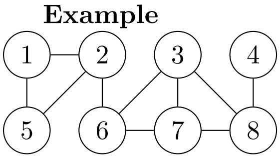


Repeat the argument on the component the get the cycle $2-6-5-2$, and attach this in to get $1-2-6-5-2-3-6-5-4-1$ as an Eulerian circuit.

Bridges

Definition
An edge of a graph is a bridge (or cut edge) if removing this edge produces a graph with more components.

Example

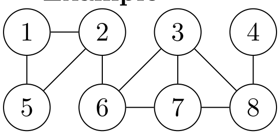


Its bridges are $(2,6)$ and $(4,8)$.

Theorem
If a graph G has m components and e is a bridge, then $G \setminus \{e\}$ has $m+1$ components.
Proof:
Let $(v,u) \in E(G)$ be a bridge. Define V to be the component of v in $G \setminus \{(v,u)\}$ and U to be the component of u in $G \setminus \{(v,u)\}$. Suppose x is in the component of v in G, but is not in V. Then because the only edge we deleted was (v,u), the path connecting v to x must have (v,u), and the remaining path shows u has a path to x, so $x \in U$. Therefore, any vertex in the original component of v and u is either

in U or V, but not both (otherwise we would connect U and V, and there is only one edge between them in G). Hence, we did not change any component of G except that of v and u's, and we made two components from one, so $G \setminus \{(v, u)\}$ has $m + 1$ components. ■

Example



Notice an edge is either contained in a cycle, or it is contained in a bridge, but not both. This generalizes:

Theorem

An edge of a graph G is a bridge if and only if it is not in a cycle.

Proof:

Let (v, u) be in a cycle, say $v - u - u_2 - \cdots - u_n - v$. Removing (v, u) leaves a graph where v has a path to u, by the cycle, and so any vertex connected to v in G is connected to u in $G \setminus \{(v, u)\}$, so deleting (v, u) does not create a new component, so it is not a bridge. Conversely, assume that (v, u) is not a bridge. Therefore, after removing it, there still exists a path from u to v, say $u - u_1 - \cdots - u_n - v$, but then in the original graph G, there is also an edge $u - v$ that completes this path to a cycle. ■

Theorem

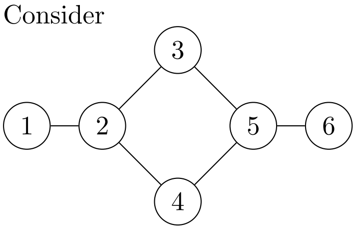
If there exist two distinct paths from u to v, then G contains a cycle.

Proof:

Take an edge that is in one path but not the other. If we remove this edge, then because we still have a path from u to v, our graph still has the same number of components, so this edge was not in a bridge, and thus was in a cycle, so our graph has a cycle. ■

Example

Consider



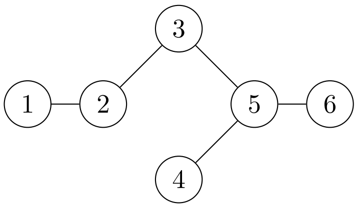
This has two distinct paths from 1 to 6, namely $1-2-3-5-6, 1-2-4-5-6$. Notice if we remove the edge $(2, 3)$, we still have a walk from 2 to 3, we go $2-1-2-4-5-6-5-3$, so there exists a path from 2 to 3 without the edge $(2, 3)$, this is $2-4-5-3$. Adding the edge we deleted back in gives a cycle $2-4-5-3-2$.

Trees

Definition

A graph is a tree if it contains no cycles and it is connected.

Example

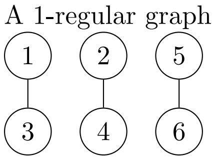


This is a tree

Definition
A graph is a forest if all of its components are tree.

Definition
Let T be a tree. A vertex $v \in V(T)$ is a leaf it has degree 1.

Example
A 1-regular graph is a forest, and every vertex is a leaf, for example,



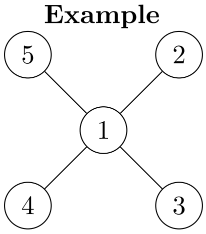
Theorem
Let T be a tree. Then there is a unique path from u to v for all $u, v \in V(T)$.
Proof:
By connectedness, there is a path, and if there were more than one path, it would imply by the last theorem of the above section that the graph would contain a cycle. ■

Theorem
Every edge of a tree is a bridge.
Proof:
The tree does not have any cycles, so its edges cannot be contained in any. ■

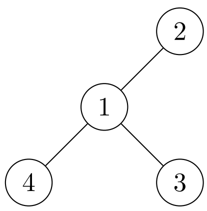
Corollary
Every edge of a forest is a bridge.
Proof:
Every edge of a forest must uniquely be contained in a tree (the component it is in), apply the above theorem to it. ■

How many edges do trees have?

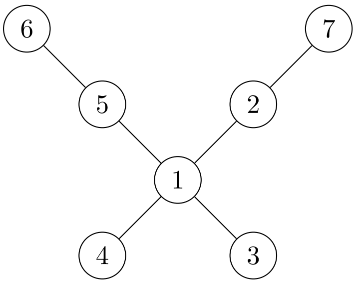
Example



has 5 vertices, 4 edges,



has 4 vertices, 3 edges, and

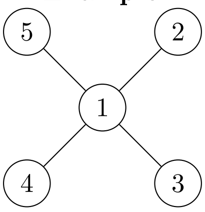


has 7 vertices, 6 edges. Does this pattern hold? Yes.

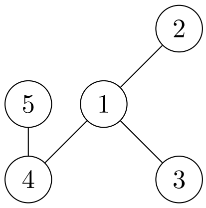
Theorem
Let T be a tree. Then $|E(T)| = |V(T)| - 1$.
Proof:
We induct on the number of vertices. The base case 1 vertex is immediate (it has no edges), and 2 vertices is true since the tree on 2 vertices is connected and hence has exactly 1 edge. Now assume this is true for all trees with $m \geq 1$ or fewer vertices, and suppose T is a tree with $m+1$ vertices. Thus T has at least 2 vertices, say u and v, and because there is a path from u to v, there is at least one edge in T. This edge is a bridge, so if we remove it, we have 2 components. Both components are trees because T is, and have m or fewer vertices because we removed something from T. Say the first of these components has s vertices, and the second has t vertices. Then $s, t \leq m$ and $s + t = m + 1$. By induction, the first tree has $s - 1$ edges, and the second tree has $t - 1$ edges. Adding the edge we removed back in, this tells us T has $(s-1) + (t-1) + 1 = s + t - 1 = m$ edges. ■

How many leaves does a tree have?

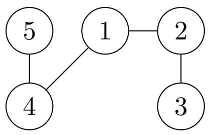
Example



has 4 leaves,



has 3 leaves,



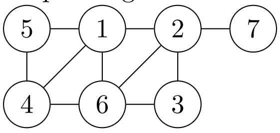
has 2 leaves. Can a tree have fewer than 2 leaves, or more than $p - 1$ leaves?

Theorem
Let T be a tree with at least 2 vertices. Then T has at least 2 leaves.
Proof:
We prove this by induction. The base case 2 vertices is 2 vertices with an edge between them (this is the only tree on 2 vertices), where both vertices are leaves. Thus inductively assume the theorem holds for every tree with $m \geq 2$ or fewer vertices. Let T be a tree with $m + 1 \geq 3$ vertices. This tree will have an edge and this edge is a bridge. Call this edge (u, v). Let U and V be the trees containing u and v that are their components in $T \setminus \{(u, v)\}$. We have two cases, either one of u and v is a leaf and the other is not, or both are not (they cannot both be leaves because $m + 1 \geq 3$). In the first case, without loss of generality suppose u is a leaf, and v is not. Then V is a tree with $m \geq 2$ vertices (U is an isolated point because u was a leaf) so by induction, V has at least two leaves, at least one of which is not v, call it w. Then u is a leaf of T (by assumption), and w is a leaf of T because it was a leaf of V and is not affected by adding the edge (u, v) back (since it will not have an edge to u because we are in a tree). Thus, T has at least 2 leaves. In the second case, suppose neither u nor v are leaves. Then, U and V both have at least two vertices (if they had a single vertex, u and v would be leaves), so by induction, both of them have at least two leaves. At least one leaf of U is not u, call it x, and at least one leaf of V is not v, call it y. Then x and y are leaves of T, similar to above. ■

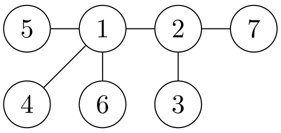
Spanning Trees

Definition
We say T is a spanning tree of a graph G if T is a tree and a spanning subgraph of G ($V(T) = V(G)$).

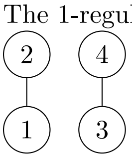
Example
A spanning tree of



is



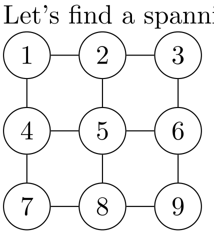
Example
The 1-regular graph on 4 vertices,



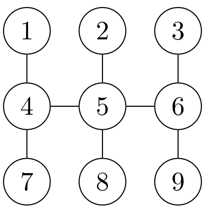
has no spanning subtree because it is disconnected.

Theorem
A graph G has a spanning tree if and only if G is connected.
Proof:
First, say G has a spanning tree T, and let $u, v \in G$. Then there is a path from u to v in T because T is a tree (and thus connected), and because T is a subgraph of G, the path uses edges in $E(G)$, giving a path from u to v in G, so G is connected. Conversely, assume G is connected. $E(G)$ contains some bridges and some edges that are not bridges. Create a subgraph of G by taking all vertices of G and all edges of G, and one-by-one removing edges that are not bridges. If there are no such edges to remove, we have a tree and we are done. Repeat this removal on the new subgraph until the process terminates, and it will terminate since $E(G)$ is finite. This process never removes a vertex, so the subgraph it creates is always spanning, and we never remove a bridge, so the subgraph it creates is always connected. This construction results in a spanning connected subgraph where every edgee is a bridge, i.e. a spanning tree. ■

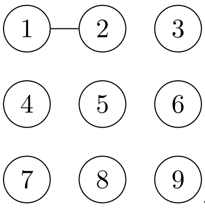
Example
Let's find a spanning tree of

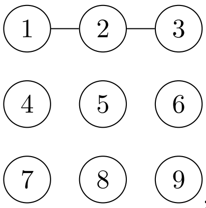


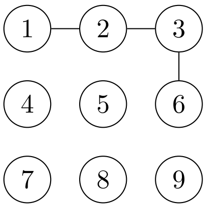
Just remove non-bridges until every edge is a bridge, in any order:

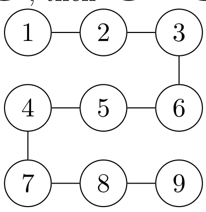


Another method that works: Start with any vertex. This will be our starting (non-spanning) tree. Find any edge from inside our tree to outside our tree. Add this edge to get a larger tree. Repeat until we have added every vertex of G. For instance on the above example, starting with (1)



, then 

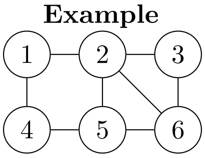
, then 

and so on, until we get something like 

Bipartite Graphs (again)

Theorem
A cycle of odd length is not bipartite.
Proof:
Assume for the sake of contradiction that a cycle G is bipartite with partition (A, B). Let v be a vertex. Assume without loss of generality that $v \in A$. The two vertices adjacent to v must be in B by bipartiteness, and the vertices adjacent to those must be in A. Continuing this process, we see vertices with a minimal path of odd length to v are in B, and vertices with a minimal path of even length to v are in A, but since G is an odd cycle, v is an odd distance from itself, showing $v \in B$, violating the fact $A \cap B = \emptyset$, a contradiction. Therefore, G is not bipartite. ■

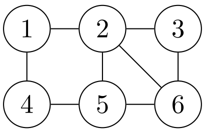
Corollary
If G contains a cycle of odd length, then it is not bipartite.
Proof:
Subgraphs of bipartite graphs must be bipartite. ■

Example


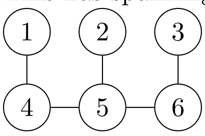
is not bipartite, because of the triangle $2-3-6-2$ (odd cycle).

Theorem
A graph is bipartite if and only if it does not contain a cycle of odd length.
Proof:
The forward direction is the above corollary, in the reverse direction, assume G is not bipartite. By looking at each component individually, we may assume without loss of generality that G is connected. We will show that G contains a cycle of odd length. Because G is connected, it has a spanning tree T. All trees are bipartite, so T has a bipartition (A, B). As G is not bipartite, there exists an edge $(u, v) \in E(G) \setminus E(T)$ such that either $u, v \in A$ or $u, v \in B$, without loss of generality say they are in A. Because T is a tree, there is a path from u to v in T, and the path is even length because T is bipartite (so it must alternate going from A to B and end in A). Connecting (u, v) to this path gives a cycle of odd length, proving the result. ■

Example
Consider the non-bipartite graph



This has spanning tree

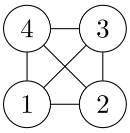


with bipartition $A = \{1, 5, 3\}$, $B = \{2, 4, 6\}$. We find an edge with both ends in A or both ends in B, for example, $(2, 6)$, and this gives our odd cycle by adjoining it with the path from 2 to 6 in our spanning tree.

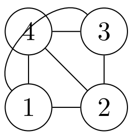
Planar Graphs

Definition
Recall that a graph is planar if we can draw the graph such that no edges cross (i.e. if we can embed it, not just immerse it, in the plane $\mathbb{R}^2$).

Example
One drawing is not enough to tell if a graph is not planar. Consider K_4 :



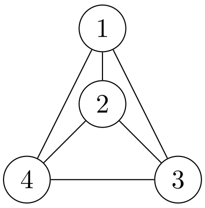
We can "stretch" the edge $(1, 3)$ out:



Example
Not every graph is planar, we will prove that K_5 does not have a planar embedding.

Definition
Let G have a planar embedding. Then a face of the embedding is an interior or exterior region of the graph (courtesy of the Jordan curve theorem).

Example
Consider the embedding



of K_4, where its four faces are the interiors of the triangles $1-2-4, 1-2-3, 2-4-3$, and the exterior of the graph. So this graph has 4 vertices, 6 edges, and 4 faces.

Example

Let T be a tree. Then T is planar (courtesy of it not having any cycles), and because it has no cycles, it has no interior regions because none of its subgraphs are cycles, so its only face is the whole plane. We see in this example, we have 1 face, $V(T)$ vertices, and $V(T)-1$ edges.

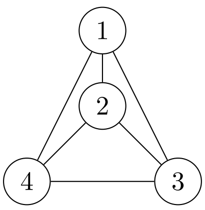
Example
Let G be a connected 2-regular graph (i.e. a cycle). Then G is planar (draw the cycle in the obvious way), and has 2 faces, $V(G)$ vertices and $V(G)$ edges (by 2-regularity).

Facts
If G is isomorphic to H, then G is planar if and only if H is planar (take the embedding that works for G, then relabel its vertices to those of H via the isomorphism).
If G is planar, then all of its subgraphs are planar (since we are never adding new edges, only possibly deleting them).

Definition
We say two faces are adjacent if they "share" an edge (i.e. the same edge is used to define their two areas).

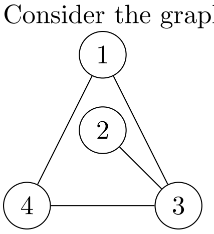
Definition
Let $u_1, \dots, u_n$ be a minimal walk containing all edges defining a face and visiting all vertices touching the face. The length of this walk is called the degree of the face.

Example
Consider the embedding of K_4



Consider the face f_1 defined by the triangle $1, 2, 3$. Then the minimal walk containing all edges is $1, 2, 3, 1$, which has length 3, so the degree of f_1 is 3, and similarly the degree of all faces of K_4 is 3.

Example
Consider the graph



This has two faces, the interior f_1 and the exterior f_2. As before, we have $\deg(f_2) = 3$, but the minimal walk containing all edges defining f_1 and visiting 2 is $1, 4, 3, 2, 3, 1$, which has length 5, so $\deg(f_1) = 5$.

In all our examples, notice that $\sum_{f \in \text{faces}} \deg(f) = 2|E(G)|$, which holds in general:

Theorem
For a planar graph G, given any planar embedding (note that the degree of any given face depends on the embedding)

$$\sum_{f \in \text{faces}} \deg(f) = 2|E(G)|$$

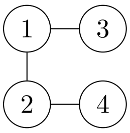
Proof:

When we add the degree of the faces, we count every edge twice, one for each side of the edge. ■

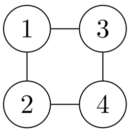
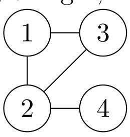
This is known as the handshake lemma for faces, akin to the handshake lemma for edges $\sum_{v\in V(G)} \deg(v) = 2|E(G)|$.

Euler's Formula

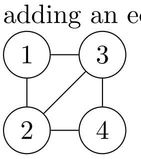
Example
Consider a connected graph with 4 vertices. What is the relation between the number of faces and the number of edges? Well any tree, like



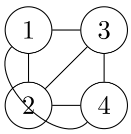
has 4 vertices, 3 edges, 1 face. If we add an edge, we have either (up to isomorphism)


or


which have 4 vertices, 4 edges, 2 faces, so we see increasing the edges by 1 increases the faces by 1. Again adding an edge, we have something like

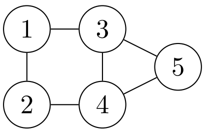


which has 4 vertices, 5 edges, 3 faces, so we increase the faces by 1 again. If we add another edge, we have something like

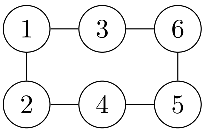


which has 4 vertices, 6 edges, 4 faces, so again, when we increase the number of edges by 1, we increase the number of faces by 1.

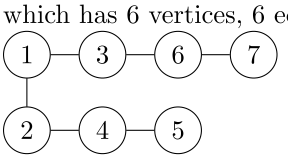
What if we increase the number of vertices, but keep the number of edges the same?



which has 5 vertices, 6 edges, and 3 faces. Adding a vertex,



which has 6 vertices, 6 edges, and 2 faces, and adding a vertex,

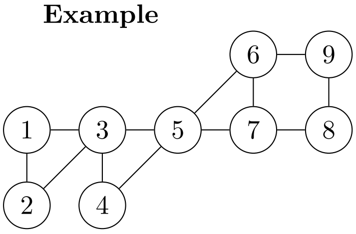


which has 6 vertices, 7 edges, and 1 face.

We note that if we increase the number of vertices, we decrease the number of faces.

Checking on any of these graphs, we see that if F is the number of faces, V is the number of vertices, and E is the number of edges, we have $V - E + F = 2$. Let's test this on a random planar graph:

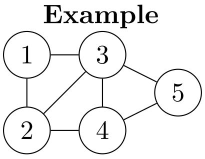
Example



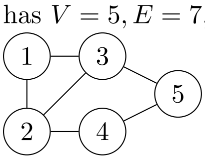
and we can check the above formula holds for this graph!

Theorem (Euler Characteristic)
Let G be a planar connected graph with V vertices, E edges and F faces. Then $V - E + F = 2$.
Proof:
We prove this by induction on the number of edges.
Let G have V vertices. The graph with a minimal number of edges that is connected is a tree, which has $E = V - 1$ edges. The only face of this planar graph is the outside face, hence $F = 1$, and we see that $V - E + F = V - (V - 1) + 1 = 2$, so the formula holds for trees, proving the base case. Now assume it holds for all $V - 1 \leq e \leq E - 1$, and G is a connected planar graph with E edges and V vertices. Notice that because of the bound on $E - 1$, G is not a tree. Hence it contains an edge that is not a bridge. The faces on either side of this edge are different, since it is not a bridge (so it is part of a cycle and thus encloses a face and has a face outside). Remove this edge to get a graph with $E' = E - 1$ edges, and still V vertices. Removing this edge merges the two faces on either side, giving that the new graph has $F' = F - 1$ faces. By induction, $V - E' + F' = 2$, implying $V - E + F = V - (E' - 1 + 1) + (F' - 1 + 1) = V - E' + F' = 2$, proving the result. ■

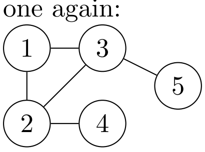
Example



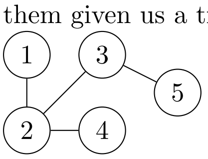
has $V = 5, E = 7, F = 4$. $(3, 4)$ is not a bridge, so removing it, we merge two faces to one:



and we see $V = 5, E = 6, F = 3$. The edge $(4, 5)$ is not a bridge, so removing it, we merge two faces to one again:



and we see $V = 5, E = 5, F = 2$. The only non-bridges we can remove are $(1, 2), (1, 3), (2, 3)$, any of them given us a tree, for example

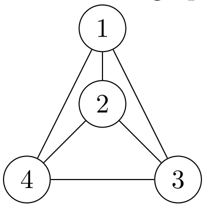


with $V = 5, E = 4, F = 1$, where our base case holds.

Question

Can we find a 3-regular planar graph where every face has degree 3? If so, what does it look like? Then by the handshaking lemmas for vertices and degrees, by the Euler characteristic, we have $|V(G)| + |F(G)| - |E(G)| = 2$, but $|V(G)| = \frac{2}{3}|E(G)|$ and $|F(G)| = \frac{2}{3}|E(G)|$, giving that $\frac{1}{3}|E(G)| = 2$, so $|E(G)| = 6$.

If such a graph exists, it thus has 6 edges, 4 vertices, and 4 faces. K_4 is an example (a tetrahedron).



Such a graph is called a Platonic solid:

Definition

A graph G is a Platonic solid if
(1) If it is connected and planar.
(2) All vertices have the same degree that is greater than or equal to 3.
(3) All faces have the same degree.

Example

We showed K_4 is a platonic solid where all vertices have degree 3 and all faces have degree 3.

Theorem

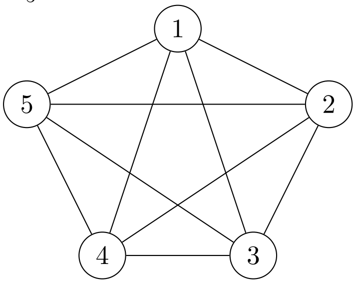
There are only 5 Platonic solids.
Proof:
Assume $\deg(f) = d_f$ and $\deg(v) = d_v$ for every vertex v and face f. Let the Platonic solid have p vertices, q edges, and f faces. By the handshake lemma, $2q = \sum_{v \in V(G)} \deg(v) = pd_v$, and $2q = fd_f$. This gives $p = \frac{2q}{d_v}$, $f = \frac{2q}{d_f}$, so by Euler's formula, we have $2 = p + f - q = \frac{2q}{d_v} + \frac{2q}{d_f} - q$, so $2 = q(\frac{2}{d_v} + \frac{2}{d_f} - 1)$, and so we have $\frac{2}{d_v} + \frac{2}{d_f} - 1 > 0$. By definition of a face, we have $d_f \geq 3$, and $d_v \geq 3$ by definition. We see that if $d_v \geq 6$ and $d_f \geq 3$, then $\frac{2}{d_v} + \frac{2}{d_f} \leq \frac{2}{6} + \frac{2}{3} - 1 = 0$, so we may assume $3 \leq d_v \leq 5$, and similarly $3 \leq d_f \leq 5$. Then we can enumerate the possible cases:
(1) If $d_v = d_f = 3$, we have $q/3 = 2$ so $q = 6$, which is K_4 we saw previously, which can be thought of as a tetrahedron.
(2) If $d_v = 3$, $d_f = 4$, then $q = 12$, $p = 8$, $f = 6$, which can be thought of as a cube.
(3) If $d_v = 4$, $d_f = 3$, then $q = 12$, $p = 6$, $f = 8$, which can be thought of as an octahedron.
(4) If $d_v = d_f = 4$, then $\frac{2}{d_v} + \frac{2}{d_f} \leq 0$, which this is impossible, which also precludes $d_v = 4, d_f = 5$, $d_v = 5, d_f = 4$, and $d_v = d_f = 5$.
(5) If $d_v = 3$, $d_f = 5$, then $q = 30$, $p = 20$, $f = 12$, which can be thought of as an dodecahedron.
(6) If $d_v = 5$, $d_f = 3$, then $q = 30, p = 12, f = 20$, which can be thought of as an icosahedron.
TODO, prove these are the same isomorphism class, after which we see that there are only 5 possible cases. ■

Fact

The planar dual of a Platonic solid is a Platonic solid: the tetrahedron is a self-dual, the octahedron and cube are duals, and the icosahedron and dodecahedron are duals.

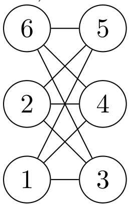
Non-Planar Graphs

Theorem
K_5 is not planar.
Proof:
K_5 is:



Assume for the sake of contradiction that K_5 is planar. We see $p = 5, q = 10$. Then by Euler's formula, we have $f = 7$. We know $\deg(F) \geq 3$ for all faces F, so by the handshaking lemma, $20 = 2q = \sum_{F \in \text{faces}} \deg(F) \geq \sum_{i=1}^{7} 3 = 21$, a contradiction. Hence K_5 is not planar. ■

Theorem
$K_{3,3}$ is not planar.
Proof:
$K_{3,3}$:



$K_{3,3}$ has 6 vertices, 9 edges, so by Euler's formula it has $2 - 6 + 9 = 5$ faces. The minimal degree of a face in a bipartite graph is 4, since cycles are of even length, which in the exact same way as above leads to the contradiction $18 \geq 20$. ■

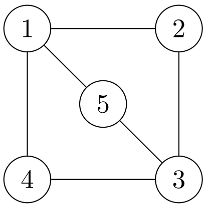
Theorem
Let G be a connected planar graph with $p \geq 3$ vertices and q edges. Then $q \leq 3p - 6$.
Proof:
As G is planar and conenected, we have $p + f - q = 2$. We further know that $\deg(f) \geq 3$ for all faces. By the handshake lemma, $2q = \sum_i \deg(f_i) \geq 3f$, which since $3p + 3f - 3q = 6$, shows $3p + 2q - 3q \geq 6$. ■

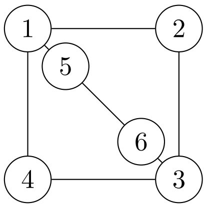
Theorem
Let G be a planar graph. Then there exists a vertex $v \in V(G)$ such that $\deg(v) \leq 5$.
Proof:
If every vertex had $\deg(v) \geq 6$, then $2q \geq 6p$ by the handshake lemma, so $3p \leq q \leq 3p - 6$ by the above result, a contradiction. ■

Kuratowski's Theorem

Definition
An edge subdivision of a graph G is done by replacing an edge with a path of length greater than or equal to 1.

Example

An edge subdivision of 

is 

Fact

Let G_2 be an edge subdivision of G_1. Then G_1 is planar if and only if G_2 is planar.

Fact

Let H be a subgraph of G. Then if G is planar, H is planar.

Theorem (Kuratowski)

A connected graph G is not planar if and only if it contains a subgraph that is an edge subdivision of K_5 or $K_{3,3}$.

Proof:

In one direction, we see an edge subdivision of $K_{3,3}$ or K_5 is not planar, and hence G is not planar. The reverse direction is really hard, which we'll skip. ■

Example

K_6 is not planar. If we remove a vertex and all edges connected to it, we just have K_5, which is not planar.

Example

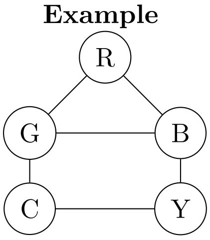
The GHMS graph on vertices $\{1, 2, 3, 4, 5, 6, 7\}$ is not planar, because it contains $K_{3,3}$ as a subgraph (bipartition even and odd vertices).

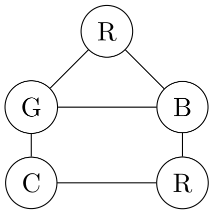
Colourings

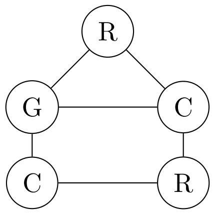
Definition

Let G be a graph. A k-colouring of the vertices of G is a map from $V(G)$ to a set of size k such that adjacent vertices are assigned different values.

Example



is a 5-colouring, 

is a 4-colouring, 

is a 3-colouring.

This graph does not have a 2-colouring:

Fact

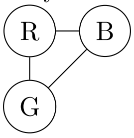
A graph is 2-colourable if and only if it is bipartite (the bipartition forms a 2-colouring).

Fact

A graph is 1-colourable if and only if it has no edges.

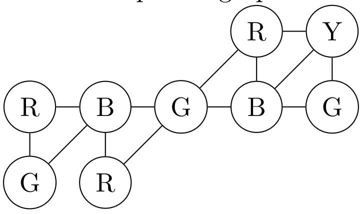
Example

A cycle of odd length is 3 colourable, e.g.:



Example

Consider a planar graph:



We see this is a 4-colouring. In general, any planar graph can be 4-coloured. One way to think of this problem is to colour different countries on a map different colours if they share a border. Something weaker:

Theorem

Let G be a planar graph. Then G is 6-colourable.

Proof:

We prove this by induction on $|V(G)|$. As a base case, the theorem holds trivially if $|V(G)| \leq 6$, so assume the statement holds for up to $p \geq 6$ vertices, and let G be a planar graph with $p+1$ vertices. As G is a planar graph, there exists $v \in V(G)$ such that $\deg(v) \leq 5$. Let G' be the graph where we remove v and all edges to v. Then G' is a planar graph with p vertices, so by induction we can 6-colour it, but because v only has (at most) 5 neighbours, we can choose a valid colour for v (a colour that is not one of its neighbours), which gives a 6-colouring of the whole graph. ■

Theorem

Let G be a planar graph. Then G is 5-colourable.

Proof:

We prove this by induction on $|V(G)|$. The base case $|V(G)| \leq 5$ holds immediately, so assume the result holds for graphs with $p \geq 5$ vertices. Let G be a planar graph with $p+1$ vertices. As G is planar, there exists a vertex v with $\deg(v) \leq 5$. If $\deg(v) \leq 4$, as before, removing v, applying the induction hypothesis to colour the remaining graph, then picking the leftover colour for v works to give our graph a 5-colouring. Otherwise, suppose $\deg(v) = 5$. Let its neighbours be labeled A, B, C, D, E. Consider the graph on A, B, C, D, E and all edges from G. This subgraph is not K_5 because G is planar, giving at least two vertices that do not have an edge between them, say A and B. Construct a new graph G' where we remove A, B, and v, and add a new vertex AB, where if $(u, v) \in E(G)$ with $v = A$ or $v = B$, then $(u, AB) \in E(G')$. Then G' is a planar graph, so by induction it has a 5-colouring, and use this to colour G. Then A and B must have the same colour by construction of G', and this is valid because there is no edge between them. This gives that A, B, C, D, E is colourable with at most 4 colours, so we can pick the remaining one to colour v, giving a 5-colouring, proving the result. ■

Theorem

Every planar graph is 4-colourable.

To prove this:

Step 1: Assume every face is a triangle.

Step 2: Break this into 633 cases.

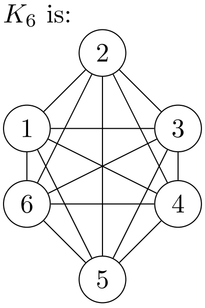
Step 3: Ask a computer for help.

Step 4: Defend yourself from mathematicians that hate computers.

Matchings

Definition
Let G be a graph. A matching M is a 1-regular subgraph of G, usually considered as a set of edges.

Example
K_6 is:



where a matching of K_6 is $(1,6),(2,5),(3,4)$. Alternatively, a nonmaximal matching is $(3,6)$.

Definition
We say a vertex v is saturated by a matching M if it is incident to an edge in M, otherwise it is unsaturated.

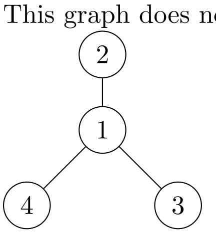
Example
In our above K_6 example, the first matching saturates every vertex, and the second matching only saturates 3 and 6.

Definition
We say a matching is perfect if every vertex is saturated.

Example
In the above K_6 example, the first matching is perfect, the second is not.

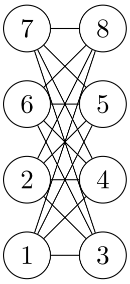
Example
K_5 does not have a perfect matching because it has an odd number of vertices.

Example
This graph does not have a perfect matching:

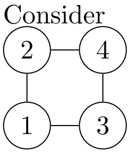


Example
Let G be a bipartite graph with bipartition A and B. If G has a perfect matching M then $|A| = |B|$. To see this, we see every edge in M is incident to a vertex in A and a vertex in B, and this reaches every vertex in both, so $|A| = |B| = |M|$.

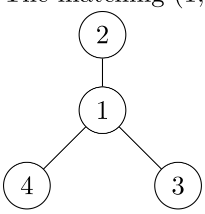
Definition
Let G be a graph with a matching M. We say a path in G is an alternating path (for M) if adjacent edges in the path have one edge in M and one edge not in M.

Example
Consider the matching of $K_{4,4}$

given by $(1,3)$, $(4,6)$, $(5,7)$. This is not perfect (as both 2 and 8 are unsaturated). Some alternating paths include 1-3 $- 2$, $2 -$ 5-7, and $8 -$ 6-4 $-$ 1-3 $- 2$.

Definition
We say an alternating path is augmenting (for M) if it starts and ends at an unsaturated vertex.

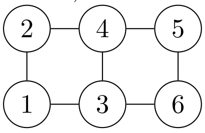
Example
Consider

with matching $(1,2)$. This has an augmenting path $4-2-1-3$. We can construct a new larger matching by removing from M every edge in M that is in the path, and add every edge not in M that is in the path, getting $(2,4)$, $(1,3)$. This process is called augmenting the matching M, and always produces a larger matching.

Definition
We say a matching M of a graph G is maximal if there are no larger matchings of G. It is easy to see that if there exists a perfect matching of G, it is maximal.

Example
The matching $(1,2)$ of

is maximal but not perfect.

Theorem
A matching M is maximal if and only if there is no augmenting path for M.
Proof:
In the forward direction, suppose contrapositively there exists an augmenting path for M. Then augmenting the path gives a larger matching than M, so M is not maximal.
Conversely, contrapositively suppose M is not maximal. Let M' be a larger matching. Consider $(M \setminus M') \cup (M' \setminus M)$ (the symmetric difference). Then this is the set of edges in exactly one of M

or M', and will create a number of paths that are all alternating. For example, consider



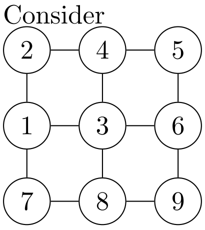
with M being $(3,4),(5,6)$, and M' being $(2,4),(1,3),(5,6)$. Then $M \setminus M'$ is $(3,4)$, and $M' \setminus M$ is $(2,4),(1,3)$, so their union is $(1,3),(3,4),(2,4)$, which we see is an augmenting path. In general, because M' is bigger than M, it will create at least one augmenting path among all of the always-alternating paths it creates. ■

Corollary

To find a maximal matching of a graph G, the following algorithm is correct:
(1) Find an augmenting path of the matching.
(2) Augment the matching (as described in the above example, remove every edge from M that is in the augmenting path, and add every edge not in M in the path).
(3) Repeat until we can no longer do (1). ■

Example

Consider



Considering the matching $(4,3),(1,7)$ with augmenting path $2,4,3,6$. Augmenting this gives $(2,4),(3,6),(1,7)$ as a matching. This has augmenting path $8,7,1,2,4,5,6,3$, where augmenting gives the matching $(7,8),(1,2),(3,6),(4,5)$, which is maximal because we have an odd number of vertices and this contains 8.

Covers

Definition

A cover of a graph G is a set $C \subseteq V(G)$ such that every edge of G has at least one end in C.

Lemma

If M is a matching of G and C is a cover, then $|E(M)| \leq |C|$.

Proof:

Because matchings are disjoint in the sense that no edges in M share a common vertex, each vertex in C accounts for at most one edge in M. ■

Lemma

If M is a matching and C is a cover of G with $|E(M)| = |C|$, then M is a maximal matching and C is a minimum-size cover.

Proof:

By the above estimate, for any matching M', $|E(M')| \leq |C|$, so if this is attained then we have a matching of maximum size. Conversely, if C' is any cover, then $|C| \geq |E(M)|$, so if we have equality then we have a cover of minimum size. ■

Theorem (König)

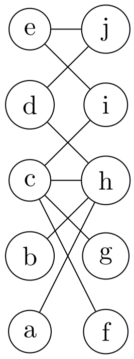
If G is a bipartite graph, then the maximum size of a matching of G equals the minimum size of a cover of G.

Proof:
Let (A, B) be a bipartition of G, and let M be a matching of G. Let X_0 be the set of vertices in A that are not saturated by M, and let Z denote the set of vertices in G that are connected to X_0 by an alternating path, where for $v \in Z$, $P(v)$ denotes this alternating path. Define $X = A \cap Z$ (the set of vertices in A with an alternating path to the unsaturated vertices in A) and $Y = B \cap Z$. Because G is bipartite, $P(v)$ has even length if $v \in X$, and has odd length if $v \in Y$. Because edges in an alternating path starting at a vertex in X_0 alternate between edges not belonging to M and edges in M (we must start with not belonging to M, so if $v \in X$, then the last edge of $P(v)$ is in M (which is vacuously true if $v \in X_0$), and if $v \in Y$, then the last edge of $P(v)$ is not in M.
We claim that $C = Y \cup (A \setminus X)$ is a cover of G.
Indeed, the only possible edges of G that are not incident to C are those from X to $B \setminus Y$ by construction of C: each side must have one end in A and one end in B, but C covers $A \setminus X$ so its A-end must be in X, and C covers Y so its B-end must be in $B \setminus Y$. But none of these exist: let $u \in X$, $v \in B \setminus Y$, and suppose $(u, v) \in E(G)$. Then we add v to the even-length alternating path $P(u)$ from X_0 to u (it is alternating by our observation above), which gives an odd-length alternating path to v from X_0, so our condition on the length above forces $v \in Y$, contradicting $v \in B \setminus Y$.
Now, we claim there is no edge of M from Y to $A \setminus X$. Indeed, let $u \in Y$, $v \in A \setminus X$, and suppose $(u, v) \in E(M)$. Then we have an odd-length alternating path $P(u)$ from X_0 to u, so appending this edge, we get an even length alternating path from X_0 to v, forcing $v \in X$ by our condition, a contradiction, so the claim holds.
Now, let U denote the set of unsaturated vertices in Y. We claim $|E(M)| = |C| - |U|$. By the previous claim, every edge of M connects either a vertex in Y to a vertex in X, or connects a vertex in $A \setminus X$ to a vertex in $B \setminus Y$. The number of edges of the first type is $|Y| - |U|$ by definition of U. For the second type, every vertex in $A \setminus X$ is saturated by M (because $X_0 \subseteq X$), so since M is matching (and cannot "double saturate"), there are $|A \setminus X|$ vertices of the second type. Thus,

$$|E(M)| = |Y| - |U| + |A \setminus X| = |C| - |U|$$

by definition of C.
Finally, note that there is an augmenting path to each vertex $v \in U$, namely $P(v)$, since it is alternating and starts in X_0 and ends in v, both of which are unsaturated. With this, the result follows quickly:
Let M be a maximal matching of G. Because it is maximal, it cannot have any augmenting paths, so the previous claim shows that $|U| = 0$. Thus, the cover $C = Y \cup (A \setminus X)$ satisfies $|C| = |E(M)|$, which by the previous lemma, gives the result (and shows C is a minimal cover). ■

Example
Consider the bipartite graph



Pick a matching $M = \{(c, g), (d, j)\}$. Then $X_0 = \{a, b, e\}$, with $X = \{a, b, d, e\}$ and $Y = \{h, i, j\}$. Take the augmenting path h, d, j, e, which gives the bigger matching $M = \{(g, c), (h, d), (e, j)\}$, which has $X_0 = \{a, b\}$, and $X = \{a, b, d, e\}$, $Y = \{g, h, i, j\}$. Take the augmenting path i, e, j, d, h, b, which gives the bigger matching $M = \{(b, h), (c, g), (e, i), (d, j)\}$ with $X_0 = \{a\}$, $X = \{a, b\}$, $Y = \{h\}$. Then, here U is empty (no vertex in Y is unsaturated by M), so the proof of the theorem shows M is maximal with

$C = Y \cup (A \setminus X)$ a minimal cover.

Definition
Let $D \subseteq V(G)$. The neighbour set of D, $N(D)$, is the set of all vertices adjacent to a vertex in D.

Theorem (Hall)
Let G be a bipartite graph with bipartition (A, B). Then G has a matching saturating every vertex in A if and only if every $D \subseteq A$ satisfies $|N(D)| \geq |D|$.
Proof:
First, suppose G has a matching M saturating every vertex in A. Then for any $D \subseteq A$, M saturates every vertex $v \in D$, and $N(D)$ contains the other end of the edge in M that is incident to v, and all of these vertices are distinct by definition of M, so $|N(D)| \geq |D|$.
Conversely, suppose contrapositively that there is no matching that saturates every vertex of A. By König's theorem, there exists a maximum matching M and a minimum cover C of G such that $|C| = |M| < |A|$, where the inequality is what we assumed above. Then, $A \cap C, A \setminus C, B \cap C, B \setminus C$ is a partition of $V(G)$. Since C is a cover, there are no edges between $A \setminus C$ and $B \setminus C$, so $N(A \setminus C) \subseteq B \cap C$. Then,

$$|N(A \setminus C)| \leq |B \cap C| = |C| - |A \cap C| < |A| - |A \cap C| = |A \setminus C|$$

so $A \setminus C$ is a subset of A that satisfies $|N(A \setminus C)| > |A \setminus C|$, as desired. ■

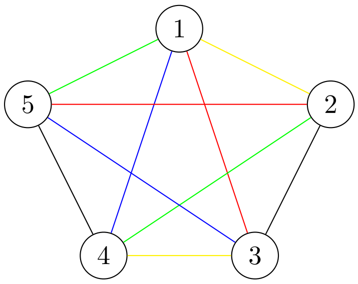
Corollary
A bipartite graph with bipartition (A, B) has a perfect matching if and only if $|A| = |B|$ and every $D \subseteq A$ satisfies $|N(D)| \geq |D|$.
Proof:
If $|A| \neq |B|$, then G has no perfect matching because all edges in G, let alone a matching, must go from A to B. On the other hand, if $|A| = |B|$, then G has a perfect matching if and only if it has a matching saturating every vertex in A, which is true if and only if for every $D \subseteq A$, we have $|N(D)| \geq |D|$ by Hall's theorem. ■

Theorem
If G is a k-regular bipartite graph for $k \geq 1$, then G has a perfect matching.
Proof:
Let (A, B) be a bipartition of G. Then every edge of M has one end in A and one end in B, so $k|A| = \sum_{v \in A} \deg(v) = \sum_{v \in B} \deg(v) = k|B|$, implying $|A| = |B|$. Let $D \subseteq A$. Since every edge incident to D has its other end in $N(D)$, we have $k|D| = \sum_{v \in D} \deg(v) \leq \sum_{v \in N(D)} \deg(v) = k|N(D)|$, so $|D| \leq |N(D)|$, and the above corollary gives a perfect matching for G. ■

Edge Colouring

Definition
A k-edge colouring of a graph G is an assignment of $1, \ldots, k$ to the edges of G such that incident edges do not have the same colour.

Example



is a 5-colouring of K_5.

Exercise
Let G be a graph that is k-edge colourable. Then $\deg(v) \le k$ for all $v \in V(G)$.

Remark
For any colour, the set of edges of that colour is a matching.

Theorem
Let G be bipartite and let $\Delta = \max_{v\in V(G)} \deg(v)$. Then G has a Δ-edge colouring. To find one, the following algorithm works:
(1) Find a matching of maximal size that saturates all $v \in V(G)$ with $\deg(v) = \Delta$.
(2) Colour the matching, remove it, and repeat.
Proof:
If the algorithm given is correct and terminates, it is easy to see it gives a valid Δ-edge colouring, and step 2 is clearly always possible provided step 1 occurs, so we need to prove step 1 is possible. Let $\Delta = \max_{v\in V(G)} \deg(v)$, and let K be all vertices with degree Δ. Let M be a matching that is maximal and saturates the maximal amount of vertices of K. If this matching saturates K we are done, so suppose there exists some vertex in K left unsaturated, and without loss of generality suppose it is in the set A in a bipartition (A, B) of G.
We perform the same construction as the proof of König's theorem. Let X_0 be the set of unsaturated vertices in $A \cap K$, and X and Y as before (the set of vertices in A with alternating paths to X_0, and the set of vertices in B with alternating paths to X_0).
We claim all $v \in X$ have degree Δ. If not, there is a vertex $v \in X$ such that $\deg(v) \ne \Delta$, with an alternating path $P(v)$ ending in X_0 of even length, where this path must have its first edge in M (see the proof of König's theorem). If we replace the edges in this path where edges in M are replaced with edges not in M and vice versa, this saturates more vertices in K than M did, which contradicts construction of M.
We claim that for every vertex $y \in Y$, there exists $x \in M$ with $(x, y) \in M$. Because M is maximal, Y every vertex in Y is saturated by M, but there are no edges in M from Y to $A \setminus X$ as seen in the proof of König's theorem, so there is an edge in M from X to y. This tells us $|Y| \le |X|$. But now, if we consider the set of edges from X to Y, we have

$$\Delta|Y| \le \Delta|X| = \sum_{v\in X} \deg(v)$$

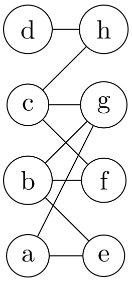
since every vertex in X has degree Δ, and

$$\sum_{v\in X} \deg(v) \overbrace{\le}^{(\dagger)} \sum_{v\in Y} \deg(y) \le \Delta|Y|$$

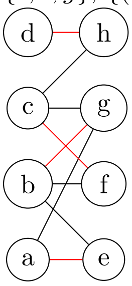
where (†) is because this sum includes all edges from Y to X, plus some from Y to $A \setminus X$, where all edges from X go to Y: they go to B by bipartiteness, but they can't be edges in M lest we saturate a vertex in X so this extends the alternating path from X to Y. Combining these, we get $|X| = |Y|$. As everything in Y has a matching to something in X, we get everything in X has a matching. Hence K is saturated. ■

Example

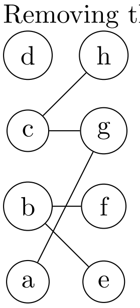
Consider the bipartite graph



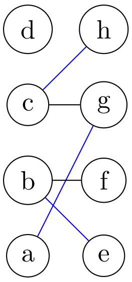
Here, $\Delta = 3$ with vertices b, c, g having degree Δ. To find a matching of maximal size that saturates $\{b, c, g\}$, $\{(a, e), (c, f), (b, g), (d, h)\}$ works, giving



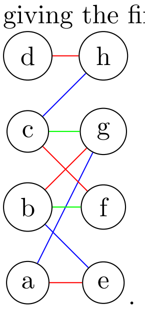
Removing these and repeating gives



This has $\Delta = 2$, which includes $\{b, c, g\}$, where a valid matching is $(a, g), (c, h)$, (b, e), giving



Removing this and repeating gives the matching $(b, f), (c, g)$, which we colour with a different colour, giving the final colouring



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This completes the course. ■